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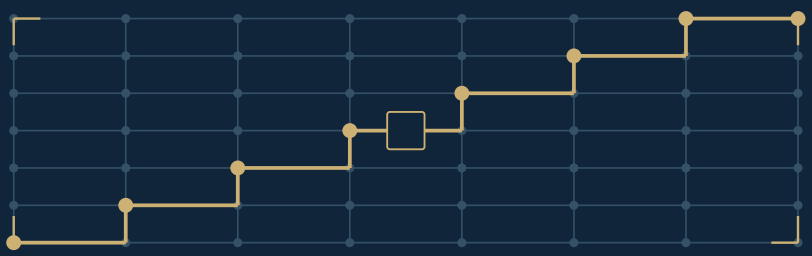
QUANTITATIVE TRADING SYSTEMS

CTRADER / SGU V2.3

Smart Grid Ultimate

Product & Support Guide

Understand the strategy. Configure with confidence.
Read the evidence in context.



H1	13	13
STUDY CHART	CURRENCY PAIRS	MONTE CARLO RECORDS

English edition / 18 October 2026

Historical simulation: January 2020–October 2026. Separate symbol tests;
bucket figures assume separately funded accounts. Trading involves risk of loss.

START HERE

A disciplined view of an automated grid

A practical guide for evaluating, configuring and operating Smart Grid Ultimate on cTrader.

Smart Grid Ultimate (SGU) is a mean-reversion cBot. It combines stochastic entry timing, optional higher-timeframe confirmation and equal-size grid legs with basket-level targets and protective controls.

The grid aims to close a group of positions when price reaches a basket target measured from its volume-weighted average entry. Persistent trends can create substantial floating losses, even when most closed trades are profitable.

01

Understand the mechanism

Signals, position management and the practical limits of protective orders.

02

Review documented evidence

Thirteen individual historical tests and thirteen matched Monte Carlo summaries.

03

Operate deliberately

Account setup, saved settings, monitoring, recovery and support.

Find what you need

What changed in the v2.2 series	03
Product mechanics and risk	04–07
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Version scope: operating guidance covers **SGU 2.3**. The preserved study uses **SGU v2.2 AB A, assembly 2.2.2.0**, labeled v2.2.2 in the workbook. These historical results were not rerun with this revision.

RELEASE HISTORY / 2.3 SIMPLIFIED EDITION

What changed in the new versions

Product improvements and maintenance fixes. Version changes do not establish future trading performance.

Retained release	Correction
v2.2.0	Improved execution, uncertain-order reconciliation and owned-position recovery; added optional Telegram alerts. Retained stochastic grid signals, spacing, sizing and targets.
v2.2.1	Corrected pip valuation for accounts in the pair's base currency, including USDCAD/USD. Preserved the monetary protection budget and existing tighter stops.
v2.2.2	Corrected cross-pair pip conversion by converting a larger arithmetic reference amount, then normalizing. Invalid conversions still block entries. Retained the prior fixes; study assembly 2.2.2.0.
v2.2.3	Corrected the optimization slowdown caused by repeated confirmed insufficient-margin refusals. New-entry retries back off through 5, 10, 20, 40 and 60 simulated minutes, capped at one hour, and material affordability improvement can release an entry sooner. Log Level = None is honored in backtesting and optimization, and repeated margin errors are aggregated.

v2.3 - simplification and correction note

Removed: the optional ADX trend gate and its five controls. The product now exposes **56 parameters**. The two stochastic indicators, optional ATR target, grid additions, basket limits, broker stops and recovery rules retain their previous behavior. RSI and the experimental signed-efficiency filter are not included.

Reason: in the examined GBPUSD H1 tick comparison, ADX Directional H4/14/25 and signed efficiency H4/14/0.30 retained the same four losing baskets as the control, reduced gains and increased equity drawdown. All three tests used USD 10,000 and 18 October 2016-17 April 2026. This is evidence about these fixed candidates and sample, not a universal claim about the indicators or a new validation of 2.3.

Retained earlier corrections: reduced repeated pip conversions and temporary allocations; consumed due margin probes; clarified Log Level = None; transient crossed quotes defer an entry and retry instead of permanently disabling entries. Existing protective exits remain active. The 2.3 protective-execution regression suite passed **126 isolated checks**; this no-ADX revision has no new native backtest.

> **Migration:** use a 2.3 preset with the five retired ADX keys removed and verify all 56 retained values. A preset that previously enabled ADX changes strategy when migrated. Historical 61-setting files and results remain archived unchanged; do not label them new 2.3 runs.

Current package: SGU Smart Grid Ultimate.algo, assembly 2.3.0.0, revision 2.3.0-no-adx.2, build no-adx-20261018.2. Sources: Product-Release / 2.3 and the archived GBPUSD filter comparison, 18 October 2026.

HOW IT WORKS

From entry signal to basket exit

One directional basket at a time within an instance, using equal-size legs. H1 means one-hour bars; D3 means three-day bars.

1

Permission to enter

The trigger seeks a stochastic cross on the latest completed H1 bar in the relevant extreme zone. The study uses 21/3/3, Wilder smoothing and 30/70 levels. D3 confirmation includes a forming-bar candidate, which can change before closure.

2

Build the ladder

After the first entry, an adverse move can add another leg at the configured distance. In the study, each leg is 0.01 lots, the step is 5 pips and the ceiling is five legs. Opposite signals do not create a second basket.

3

Manage the combined position

The target is measured from the basket's volume-weighted average entry. The study uses Average True Range (ATR), H1(14), multiplier 1, bounded to 2–30 pips. It refreshes from the last completed ATR bar when a new bar starts and can change during a basket.

4

Request and verify closure

A target, loss trigger or relevant execution event can start liquidation. Closing individual positions takes time; fills, broker responses and any remaining exposure must be reconciled.

> **Study exposure illustration:** 0.01 lots × 5 legs = 0.05 lots of intended maximum volume. The first-to-last entry span is approximately 5 pips × (5 − 1) = 20 pips, before fill differences. Broker volume normalization can change actual size.

Entry signals and basket management

The signal controls whether the first leg may open. Once exposure exists, the basket's state, price distance and protection govern management. Entry filters do not remove the loss potential of an existing basket.

The basket target is managed by the running robot; entries request a broker stop, without a separate broker take-profit. A broker-stop event, a manual leg closure or an unexpected partial-volume change can trigger liquidation of the remaining basket. Manual intervention changes the managed state.

Indicator key: stochastic %K is the oscillator line; %D is its smoothed signal line. ATR measures price-range volatility. SGU 2.3 has no ADX or signed-efficiency trend gate; the archived study had ADX disabled.

RISK ARCHITECTURE

Protective controls need execution

A configured threshold starts a response. The final loss depends on market and broker execution.

Control	Study value	Practical meaning
Money budget	USD 700 per basket	The tested allocation monitors basket loss. Reaching it requests liquidation; it is not a guaranteed final loss ceiling.
Balance-based basket stop	30% of current balance	A separate floating-loss trigger. Its monetary amount changes with balance. It does not cap historical equity drawdown at 30%.
Broker leg stop	Automatic (parameter = 0)	Zero requests budget-derived server-side protection. It does not mean “no stop.” Broker minimum distances and acceptance still apply.
Optional distance stop	Disabled (0 pips)	The additional basket stop in pips is disabled in the study. Targets and the other enabled protections remain relevant.
Exposure ceiling	Five equal-size legs	This limits intended position count. Once the last leg is open, price can continue moving against the basket.

Why a smooth balance line can mislead: balance records realized outcomes; equity also reflects open losses. A grid can accumulate many small wins while carrying a much larger floating loss. Evaluate equity drawdown as well as profit factor and win rate.

Three different thresholds: at a USD 1,000 balance, the 30% basket trigger starts at USD 300. The USD 700 money allocation is a separate basket protection input. The Monte Carlo USD 700 floor means remaining simulated equity, equivalent to a USD 300 decline from the initial capital. None is an account-wide loss guarantee.

Assess the account as a whole

Separate symbols can move together. Several cBot instances can therefore draw down at the same time. These individual tests do not establish a suitable account allocation or a combined portfolio drawdown.

Gaps, conversion changes, commission, swap, rejected orders, unavailable liquidity and a disconnected or stopped process can take realized losses beyond the intended thresholds. Keep broker-confirmed positions and stops under review.

RISK EDUCATION / DRAWDOWN AND RECOVERY

After a loss, recovery needs a larger gain

Compare the loss with the return required on the capital that remains. Educational arithmetic, not a return forecast.

In SGU: Basket Stop (% balance, 0 = off) (**StopLossPercent**) requests a basket exit when its net floating loss reaches that share of current account balance. The study used 30%; the product default is 0 (off). This is separate from **Max Basket Loss (money)** and is not an account-wide maximum-drawdown setting.

Example base: USD 1,000. Assume the loss is fully realized at exactly the percentage shown, with no other positions, deposits or withdrawals. The base can be starting capital or a previous equity peak. The same recovery percentages apply at any positive capital base.

Actual loss / drawdown	Capital remaining (USD)	Net profit needed to regain USD 1,000	Required return on remaining capital
5%	950.00	50.00	5.26%
10%	900.00	100.00	11.11%
15%	850.00	150.00	17.65%
20%	800.00	200.00	25.00%
25%	750.00	250.00	33.33%
30%	700.00	300.00	42.86%
40%	600.00	400.00	66.67%
50%	500.00	500.00	100.00%
60%	400.00	600.00	150.00%
70%	300.00	700.00	233.33%
75%	250.00	750.00	300.00%
80%	200.00	800.00	400.00%
90%	100.00	900.00	900.00%
95%	50.00	950.00	1,900.00%
100%	0.00	1,000.00	No finite return

Formula: remaining capital = base x (1 - d); required net profit = base x d; recovery return = d / (1 - d), where d is the loss as a decimal (30% = 0.30). Multiply the last result by 100 for %. Returns are net of trading costs; percentages are rounded to two decimals.

> **30% is not recovered by earning 30%.** USD 1,000 falls to USD 700; a 30% gain then reaches only USD 910. Regaining USD 1,000 needs USD 300 net profit, or **42.86%** on USD 700. Two consecutive 30% losses leave USD 490 and require **104.08%** to recover. At 100% loss, no trading return on zero capital can restore the base.

Use the actual final loss. Other active protections may close earlier; gaps, costs and execution delays can make losses larger. Multiple baskets can lose together or in sequence. This table gives no recovery time or probability. Adding outside capital is a deposit, not trading performance. Educational addition: 17 October 2026.

OPERATING REQUIREMENTS

Know the instance you are running

Ownership, account type and platform availability are part of correct operation.

Requirement	Operator action
Hedged account	Use a hedging account that preserves separate grid legs. The robot rejects netted accounts.
Desktop execution	Use a local cTrader Windows or Mac instance for new entries. This SGU implementation treats Cloud operation as exit-only and attempts to liquidate owned exposure.
Exact ownership identity	Positions belong to an exact symbol and Instance Tag. Preserve the original identity for intended recovery; use distinct tags for separate books.
Continuous availability	Keep the desktop application, account connection and operating environment available for management. A VPS is useful only if it runs the intended local desktop setup reliably.
One owner per book	Use one owner for each account, symbol and tag. The ownership lock is local to one device; check other computers and VPSs yourself. A renamed file does not create a new identity.
Verified package	Use SGU Smart Grid Ultimate.algo, assembly 2.3.0.0 / revision 2.3.0-no-adx.2. Verify build no-adx-20261018.2; legacy SGU v2.2 names and AB A study telemetry do not identify the current package.

Stopping and restarting

Normal **Stop** attempts to cancel owned orders and liquidate owned positions. It is a liquidation request, not a pause that preserves the basket. Inspect broker-confirmed results and plan upgrades while flat whenever possible.

Recovery records are local to the device. Moving a preset to another machine does not transfer the journal. A recovered active basket retains its original money allocation; changing the preset does not reset that budget.

> If an order outcome is uncertain, do not repeatedly restart, erase recovery records or change the Instance Tag to force a new entry. First reconcile the account's actual positions and orders, then resolve the recorded recovery condition.

HISTORICAL STUDY

The tested configuration

Requested dates: 1 January 2020–14 October 2026, UTC. Each symbol was tested separately.

Study item	Recorded setting
Study build	SGU v2.2 AB A · assembly 2.2.2.0 · workbook label SGU v2.2.2
Broker / account	Raw Trading Ltd · USD deposit currency · hedging · 1:20 account leverage; this sets margin availability, not a return multiplier.
Data and costs	Server tick data; commission USD 30 per million USD volume. Native spread field is unreported.
Starting capital	USD 1,000 in all thirteen actual backtests, including the replacement USDJPY run. No proportional rescaling is used.
Chart / trigger / confirmation	H1 chart and stochastic trigger; D3 stochastic confirmation. Both oscillators 21/3/3, Wilder smoothing, 30/70 levels; confirmation window four bars.
Grid / execution	0.01 lots per leg; 5-pip spacing; five legs; 2-pip maximum spread guard; slippage setting 0; added-leg guards off.
Targets / protections	ATR H1(14), Exponential, multiplier 1, range 2–30 pips; fixed fallback 30 pips. USD 700 basket allocation, 30% balance stop, automatic broker leg stop.
Entry windows	Friday 20:00–24:00 UTC and Sunday 00:00–24:00 UTC blocked. ADX filter disabled.
Monte Carlo	BlockBootstrap; 1,000 iterations; seed 0; five-trade blocks; absolute equity floor USD 700 for all thirteen saved simulations.

Evidence qualifications: all thirteen rows in the completed workbook are included. 8 runs contain logged EntityNotFound/stop-modification errors, although all exporters finished. USDJPY, EURCHF, EURCAD and CADCHF Monte Carlo summaries were calculated with the local cTrader engine. See page 61.

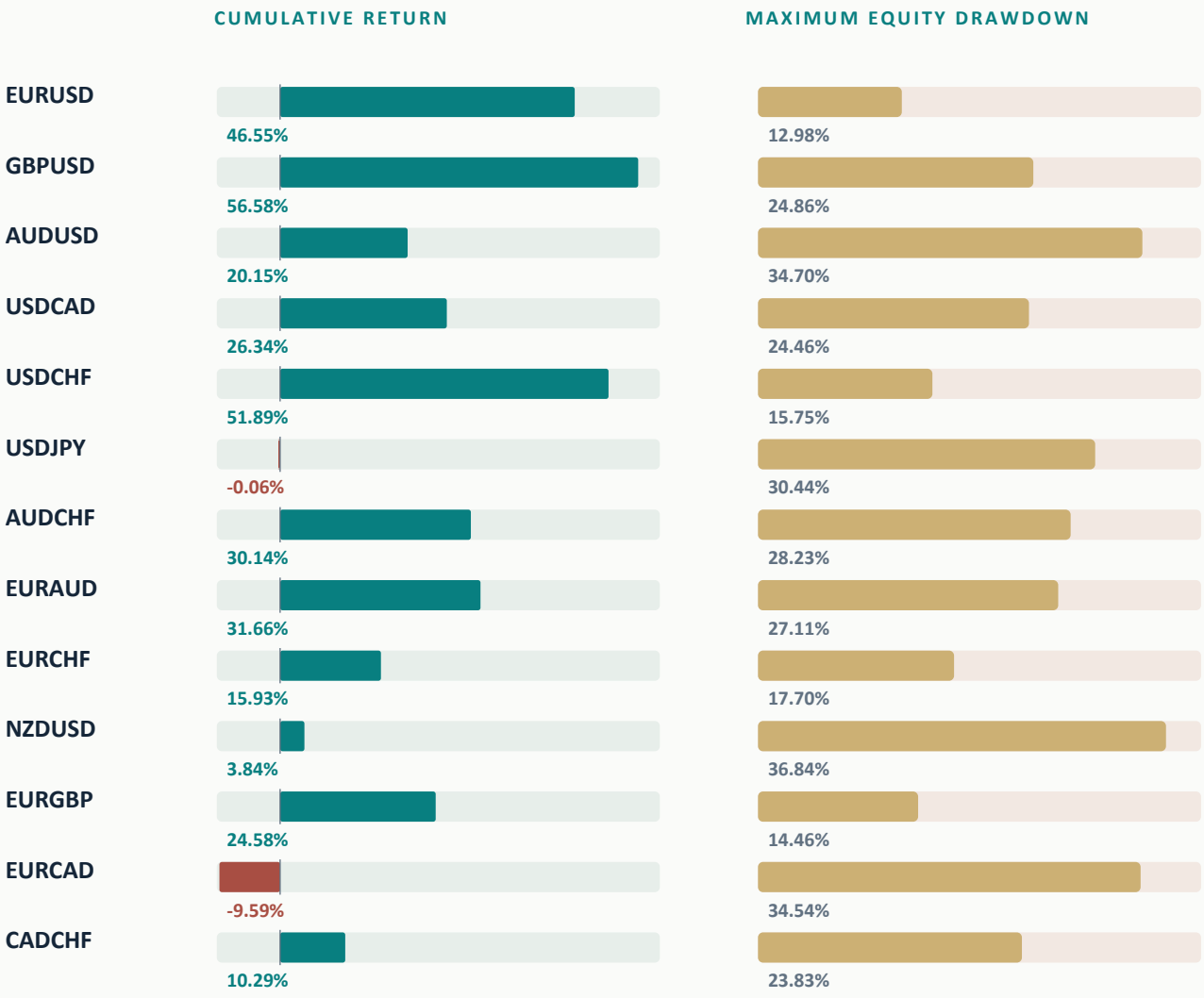
The publisher reports **no indicator fine-tuning for this study**. All thirteen saved files use the same indicator parameters. This is a common-setting, fixed-volume baseline; it is not an established optimum or a forecast of improvements from optimization. Position sizing and leverage scenarios: pages 45–47.

STUDY AT A GLANCE

Return and floating risk belong together

Reported cumulative return and maximum historical equity drawdown. Percentages are not annualized.

INDIVIDUAL TESTS	MATCHED MC RUNS	DEEPEST EQUITY DD
13	13	36.84%
3,908 closed trades in total	Common USD 700 equity floor	NZDUSD historical test



Read across each row, not across the two bar scales: return uses a −10% to +60% range; drawdown uses 0% to 40%. GBPUSD has the highest cumulative return, while EURUSD combines lower return with smaller equity drawdown. NZDUSD's 3.84% gain sits beside 36.84% drawdown. All runs start at USD 1,000 with 0.01-lot legs; these are separate tests.

11 of 13 runs ended profitable. USDJPY and EURCAD ended with losses. This observed sample does not establish an expected return or future win probability.

HISTORICAL RESULTS

All thirteen historical results

USD figures. Net profit includes the reported trading costs; return is the native rounded cumulative figure.

Pair	Initial capital	Net profit	Final equity	Return	Equity DD
EURUSD	1,000	465.49	1,465.49	46.55%	12.98%
GBPUSD	1,000	565.79	1,565.79	56.58%	24.86%
AUDUSD	1,000	201.46	1,201.46	20.15%	34.70%
USDCAD	1,000	263.41	1,263.41	26.34%	24.46%
USDCHF	1,000	518.93	1,518.93	51.89%	15.75%
USDJPY	1,000	-0.61	999.39	-0.06%	30.44%
AUDCHF	1,000	301.42	1,301.42	30.14%	28.23%
EURAUD	1,000	316.63	1,316.63	31.66%	27.11%
EURCHF	1,000	159.35	1,159.35	15.93%	17.70%
NZDUSD	1,000	38.40	1,038.40	3.84%	36.84%
EURGBP	1,000	245.75	1,245.75	24.58%	14.46%
EURCAD	1,000	-95.90	904.10	-9.59%	34.54%
CADCHF	1,000	102.91	1,102.91	10.29%	23.83%

Loss-bearing runs: USDJPY finished at USD 999.39 (net −USD 0.61; native rounded return −0.06%). EURCAD finished at USD 904.10 (net −USD 95.90; return −9.59%). Both are included in the full panel.

Read return alongside exposure

The capital base, leg size and grid settings are common across all thirteen runs. Differences in pip value, trading sequence and market history still produce different equity risk. No sum of these isolated profits is presented as a jointly traded portfolio return.

HISTORICAL RESULTS / TRADES AND LOGS

Closed trades and execution context

Win rates and profit factors describe closed legs; the last column preserves critical execution-log counts.

Pair	Trades	Win rate	Profit factor	Balance DD	Critical messages
EURUSD	339	95.28%	56.02	0.47%	0
GBPUSD	347	96.83%	164.52	0.09%	0
AUDUSD	454	92.29%	1.65	30.01%	0
USDCAD	331	96.07%	5.78	3.49%	86
USDCHF	355	94.37%	59.97	0.22%	39
USDJPY	322	95.65%	1.00	30.00%	128
AUDCHF	277	90.25%	4.18	6.02%	6
EURAUD	305	94.43%	3.86	6.62%	9
EURCHF	267	88.39%	2.04	10.26%	4
NZDUSD	317	91.48%	1.12	30.01%	0
EURGBP	229	87.34%	23.02	0.10%	5
EURCAD	211	92.89%	0.72	30.03%	1
CADCHF	154	90.26%	2.15	7.35%	0

A closed trade is a grid leg, not a complete basket. High win rates and profit factors can coexist with substantial floating losses. Critical EntityNotFound messages require execution review; a finished exporter is not proof of error-free operation.

End-of-run baskets: AUDCHF, EURAUD, EURCHF, CADCHF each contain a basket classified EndOfRun. Its P/L remains in total profit, but completed-basket win rates, averages and worst outcomes exclude it. Final position and order counts are recorded separately.

Across the thirteen logs: 278 critical messages in 8 runs. Counts identify preserved diagnostics, not a count of distinct financial losses.

RESAMPLING EVIDENCE

What Monte Carlo tells you

BlockBootstrap rearranges observed closed-trade outcomes. It does not replay new market prices.

The study resamples blocks of five closed trades over 1,000 iterations. The table reports the lower fifth percentile (P5), median and upper 95th percentile (P95) of simulated net profit, plus sample frequencies. These are full-test-horizon figures.

Pair	P5 profit USD	Median USD	P95 profit USD	Profitable	Floor USD	Breach
EURUSD	423.55	465.99	508.87	100.0%	700	0.0%
GBPUSD	523.74	565.35	609.46	100.0%	700	0.0%
AUDUSD	-272.83	259.09	515.61	82.8%	700	9.5%
USDCAD	181.26	268.21	325.80	100.0%	700	0.0%
USDCHF	469.47	518.81	572.97	100.0%	700	0.0%
USDJPY	-664.39	45.74	447.74	55.4%	700	32.8%
AUDCHF	168.47	313.09	422.62	99.5%	700	0.0%
EURAUD	161.86	331.34	442.87	99.9%	700	0.0%
EURCHF	-35.45	180.22	318.06	91.0%	700	0.8%
NZDUSD	-402.35	87.20	349.68	63.1%	700	15.1%
EURGBP	218.08	244.43	274.87	100.0%	700	0.0%
EURCAD	-545.34	-63.68	234.55	39.7%	700	27.2%
CADCHF	-38.64	113.46	201.52	89.6%	700	0.0%

One common threshold: all thirteen saved simulations use a USD 700 equity floor on USD 1,000 starting equity. USDJPY, EURCHF, EURCAD and CADCHF summaries were calculated using the local cTrader engine; the other nine are original interface exports. All source contexts were checked.

Keep two drawdowns separate

Historical equity drawdown includes sampled open-position losses. Closed-trade resampling omits the floating path inside a basket and can show much smaller drawdown. It cannot replace historical equity risk.

USDJPY: 32.8% of resampled paths breached the floor. **NZDUSD:** 36.84% historical equity drawdown was the deepest in the panel. Resampling can generate negative simulated equity and does not model broker stop-out.

An absolute equity floor is distinct from the basket loss allocation. Zero observed breaches or a high profitable frequency is not a guarantee of future performance.

PERFORMANCE METRICS / 1 OF 2

Measure growth, drawdown and time at risk

Descriptive measurements of the original thirteen tests. Annualized growth summarizes endpoints; volume stayed fixed.

Pair	Ann. growth	Equity DD	Growth / DD	Time in market	P95 hold days	Longest days
EURUSD	5.79%	12.98%	0.45	5.86%	5.84	31.41
GBPUSD	6.83%	24.86%	0.27	6.49%	6.87	46.86
AUDUSD	2.74%	34.70%	0.08	7.91%	6.31	29.56
USDCAD	3.51%	24.46%	0.14	9.46%	4.87	135.92
USDCHF	6.35%	15.75%	0.40	6.93%	5.05	36.79
USDJPY	-0.01%	30.44%	-0.0003	14.60%	8.45	239.47
AUDCHF	3.96%	28.23%	0.14	6.47%	6.68	54.42
EURAUD	4.14%	27.11%	0.15	13.40%	11.10	84.03
EURCHF	2.20%	17.70%	0.12	9.60%	10.40	103.21
NZDUSD	0.56%	36.84%	0.02	7.13%	6.71	51.73
EURGBP	3.29%	14.46%	0.23	4.21%	5.23	27.09
EURCAD	-1.47%	34.54%	-0.04	22.22%	21.42	236.80
CADCHF	1.45%	23.83%	0.06	25.57%	9.91	559.33

Growth / DD divides annualized endpoint growth by the maximum equity-drawdown percentage over this full study. It compares reward with the observed peak-to-trough risk, without assuming steady yearly gains. It is not a conventional fixed-window Calmar ratio, forecast or recommended risk budget.

Time in market is the share of the recorded run's calendar time with at least one open position, including weekends. **P95 hold** is the nearest-rank 95th percentile of basket lifetimes, from first entry to final exit. These measure exposure and waiting time, not margin use or future recovery speed.

> **Read the tail, not just the typical trade.** CADCHF's median basket lasted only 0.20 days, but its longest lasted 559.33 days and ended at shutdown. A low exposure percentage or short median can conceal a long, loss-bearing basket. Shutdown durations stop at the test endpoint.

Sharpe, Sortino, daily equity volatility and exact time underwater are not inferred from interval minima/maxima. They require aligned point-in-time equity observations. Full definitions, source checks and unrounded values: **analysis-supplement / performance-metrics.csv** and **performance-metrics-notes.md**.

PERFORMANCE METRICS / 2 OF 2

Measure the complete basket outcome

USD net of recorded commission and swap. All fully liquidated baskets, including the four final shutdowns, are counted here.

Pair	Count	Win rate	Mean net USD	Avg win USD	Avg loss USD	Payoff	Basket PF
EURUSD	115	99.13%	4.05	4.13	-5.15	0.80	91.39
GBPUSD	105	100.00%	5.39	5.39	n/a	n/a	n/a
AUDUSD	153	99.35%	1.32	3.31	-301.57	0.01	1.67
USDCAD	113	98.23%	2.33	2.87	-27.48	0.10	5.79
USDCHF	125	99.20%	4.15	4.19	-0.74	5.66	702.26
USDJPY	102	98.04%	-0.01	4.27	-213.80	0.02	0.999
AUDCHF	104	97.12%	2.90	3.84	-28.84	0.13	4.48
EURAUD	101	97.03%	3.13	4.34	-36.30	0.12	3.91
EURCHF	92	97.83%	1.73	3.40	-73.36	0.05	2.09
NZDUSD	111	97.30%	0.35	3.20	-102.29	0.03	1.13
EURGBP	81	100.00%	3.03	3.03	n/a	n/a	n/a
EURCAD	62	96.77%	-1.55	4.08	-170.24	0.02	0.72
CADCHF	56	98.21%	1.84	3.46	-87.49	0.04	2.18

Mean net (historical expectancy) = total net profit / number of closed baskets. **Payoff** = average winning basket / absolute average losing basket. **Basket PF** = total positive basket net / absolute total negative basket net. Each basket groups its grid legs; these figures differ from the individual-leg statistics on page 11.

> **A high win rate can hide an asymmetric loss.** AUDUSD won 152 of 153 baskets, but its USD 301.57 loss offset about 91 average winning baskets. Its 99.35% basket win rate therefore produced only USD 1.32 mean net per basket. The frequency of wins must be read alongside their size and the floating equity path.

n/a means no negative basket was observed. GBPUSD and EURGBP have no sample average loss, finite basket PF or payoff estimate. This is not evidence of zero future loss. Only 21 of 1,320 liquidated baskets were negative across the thirteen runs; loss-dependent ratios are unstable, and baskets are not independent observations.

These all-closed figures include EndOfRun liquidations in AUDCHF, EURAUD, EURCHF and CADCHF, so aggregate basket net equals full reported profit. The robot's **completed-basket** statistics exclude those shutdowns. Both scopes are supplied in the CSV; compare like with like.

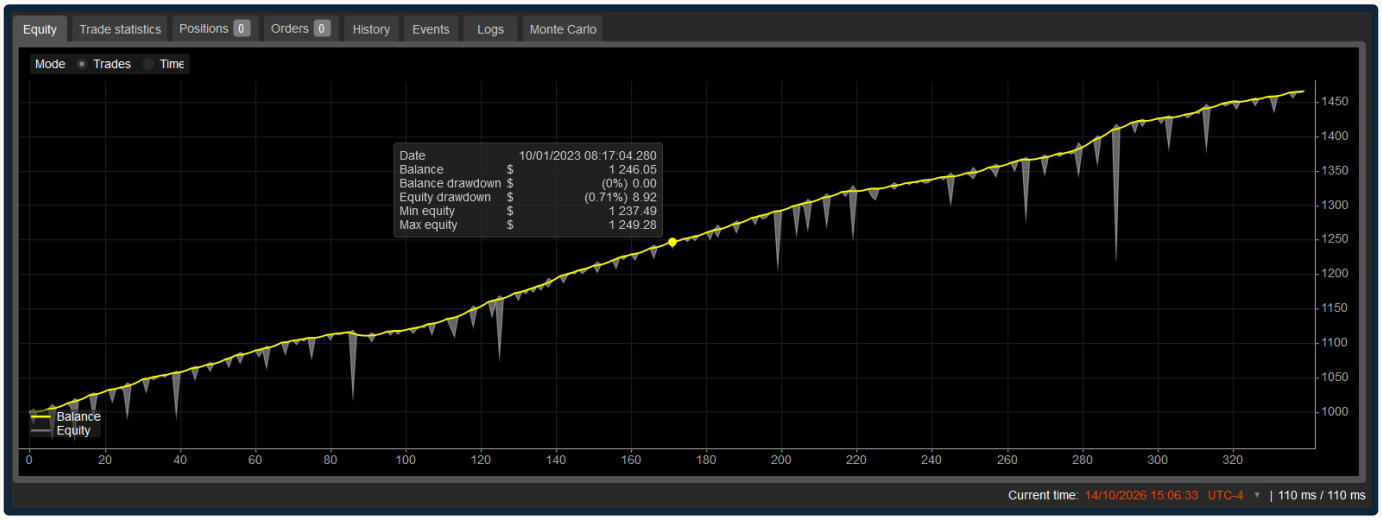
PAIR PROFILE

EURUSD

H1 · 1 January 2020–14 October 2026 · Historical simulation

NET PROFIT	CUMULATIVE RETURN	MAX. EQUITY DD
+$465.49	46.55%	12.98%

HISTORICAL BALANCE AND EQUITY



cTrader capture: EURUSD.PNG. Yellow = balance; grey = floating equity. Axis = closed trades; the dated discussion uses native timestamps.

The rising yellow balance line ends at USD 1,465.49, while repeated grey dips show open baskets carrying losses before closing. The largest equity decline was 12.98%: from a sampled peak of USD 1,409.77 on 10 April 2025 to USD 1,226.72 on 21 April 2025. A five-leg sell basket faced an adverse move during the April trade-policy repricing described on pages 41–42.

That basket eventually closed at +USD 10.17 on 12 May. The old equity peak was first touched in a sampled interval on 9 May, before the basket closed. This separates temporary equity recovery from realized profit. Balance drawdown was only 0.47%; that small closed-loss figure should not obscure the much larger floating exposure.

Basket outcome and time at risk	Archived measurement
All closed baskets / mean net	115 / $4.05
Median / P95 / longest holding time	0.12 / 5.84 / 31.41 calendar days
Time in market / baskets reaching five legs	5.86% / 33.04%

Initial $1,000.00 · final equity $1,465.49 · 339 closed legs · leg win rate 95.28% · leg PF 56.02.

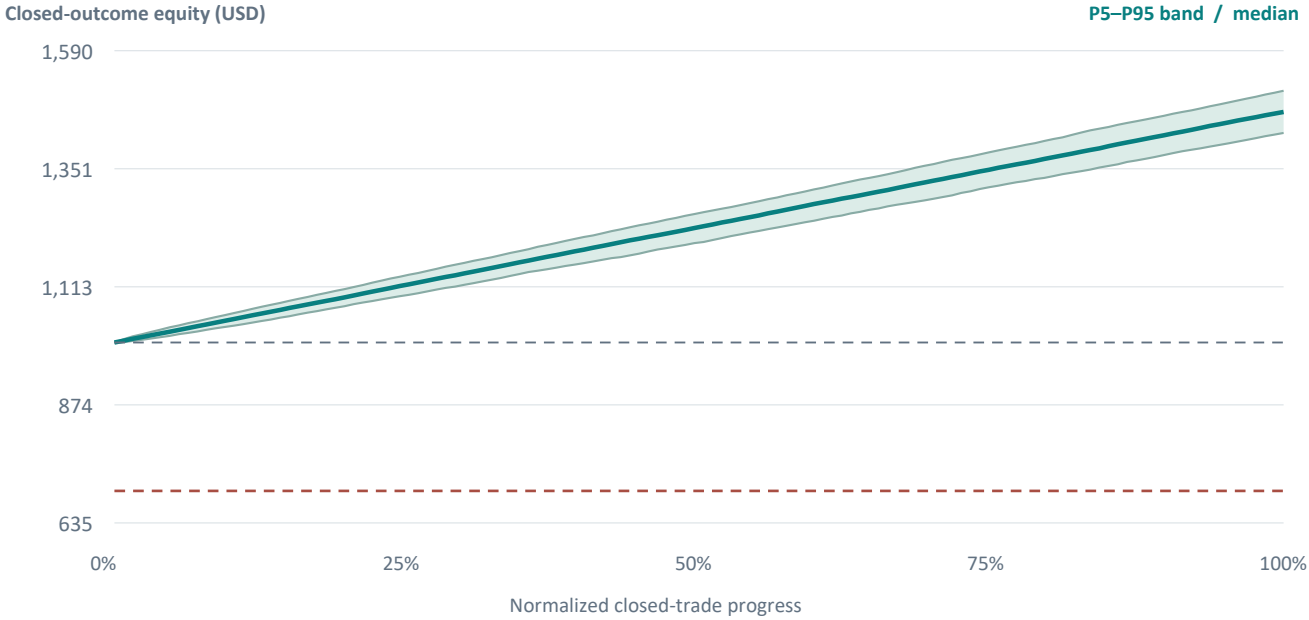
All-closed basket measurements above include shutdowns. One isolated historical test; run ID: f5ef540d74aa. Calendar exposure is not a margin or leverage measurement.

PAIR ANALYSIS / MONTE CARLO

EURUSD · resampled outcomes

Matched closed-trade BlockBootstrap · 1,000 iterations · Five-trade blocks · Seed 0

P5 NET PROFIT	MEDIAN NET PROFIT	P95 NET PROFIT
$423.55	$465.99	$508.87



Teal line = median; shaded band = pointwise P5–P95 equity; grey dashed line = USD 1,000 start; red dashed line = USD 700 floor. The 128 plotted samples are interpolated over trade progress, not future calendar dates.

The narrow, rising envelope reflects a closed-trade sample dominated by small gains. Full-horizon simulated net profit runs from USD 423.55 at P5 to USD 508.87 at P95, with a USD 465.99 median. All 1,000 resampled paths ended profitable and none breached the USD 700 floor in this calculation.

Those observations describe rearrangements of the archived outcomes. They do not establish a 100% future win probability. The 0.50% P95 resampled closed-trade drawdown is far below the 12.98% historical equity drawdown because floating losses inside baskets are absent. The smooth median should therefore be read alongside the historical chart.

MC profit: P5 $423.55 · median $465.99 · P95 $508.87. Profitable 100.0% · USD 700 floor breach 0.0%.

MC statistics and plotted band: matched original cTrader interface JSON export for this run.

A pointwise band does not give a 90% chance that an entire future path stays inside it. Floor breaches are finite-model frequencies, not account-ruin probabilities. Closed-trade resampling omits intra-basket floating losses and broker stop-out.

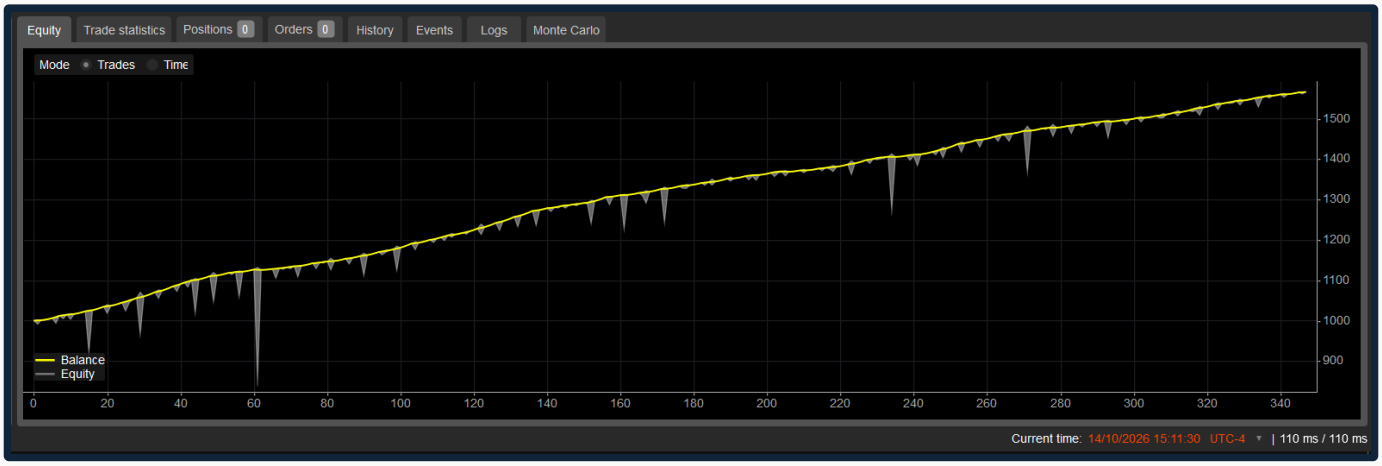
PAIR PROFILE

GBPUSD

H1 · 1 January 2020–14 October 2026 · Historical simulation

NET PROFIT	CUMULATIVE RETURN	MAX. EQUITY DD
+$565.79	56.58%	24.86%

HISTORICAL BALANCE AND EQUITY



cTrader capture: GBPUSD.PNG. Yellow = balance; grey = floating equity. Axis = closed trades; the dated discussion uses native timestamps.

Frequent profitable closes produce a steadily rising yellow balance line, ending at USD 1,565.79. The grey equity path reveals a much less comfortable episode in February 2021. Its sampled peak of USD 1,131.44 on 8 February fell to USD 850.19 on 24 February 2021, a 24.86% equity drawdown, while a five-leg sell basket was exposed.

The basket closed at only +USD 1.34 on 24 March. A profitable exit did not erase the time spent below the old account peak: its first sampled recovery touch occurred on 15 June. Balance drawdown was just 0.09%. The contrast shows why the attractive final return and very small realized drawdown cannot substitute for an assessment of open-position risk.

Basket outcome and time at risk	Archived measurement
All closed baskets / mean net	105 / $5.39
Median / P95 / longest holding time	0.18 / 6.87 / 46.86 calendar days
Time in market / baskets reaching five legs	6.49% / 40.95%

Initial $1,000.00 · final equity $1,565.79 · 347 closed legs · leg win rate 96.83% · leg PF 164.52.

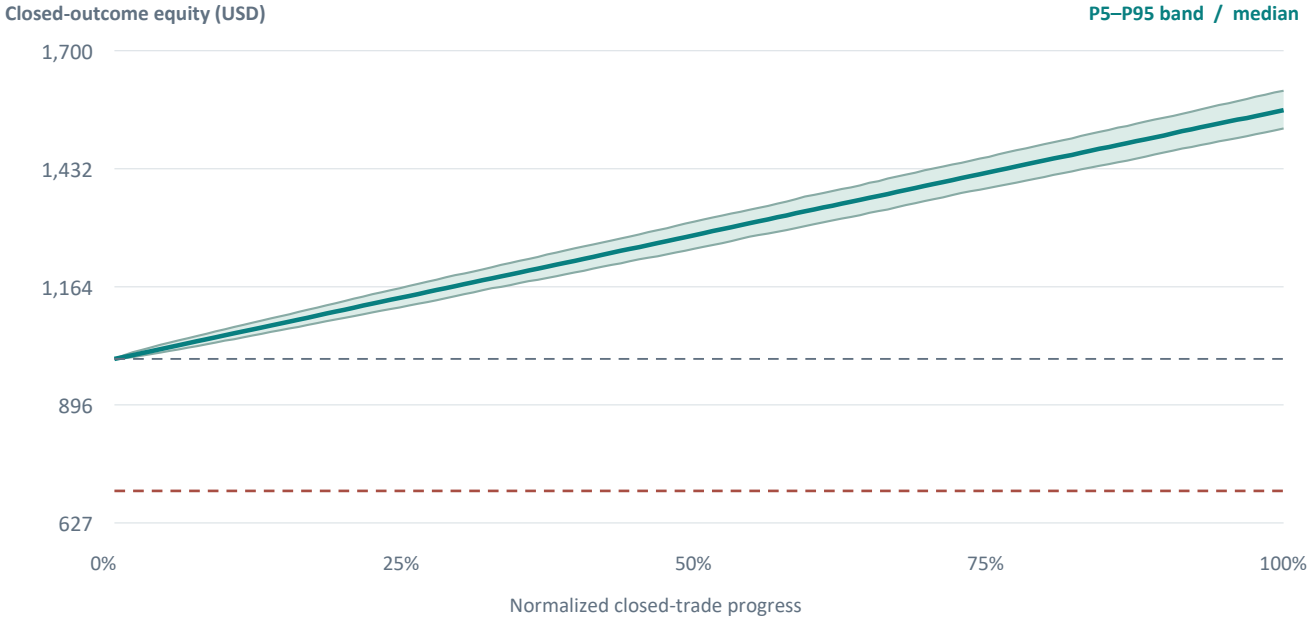
All-closed basket measurements above include shutdowns. One isolated historical test; run ID: a9a425a6da69. Calendar exposure is not a margin or leverage measurement.

PAIR ANALYSIS / MONTE CARLO

GBPUSD · resampled outcomes

Matched closed-trade BlockBootstrap · 1,000 iterations · Five-trade blocks · Seed 0

P5 NET PROFIT	MEDIAN NET PROFIT	P95 NET PROFIT
$523.74	$565.35	$609.46



Teal line = median; shaded band = pointwise P5–P95 equity; grey dashed line = USD 1,000 start; red dashed line = USD 700 floor. The 128 plotted samples are interpolated over trade progress, not future calendar dates.

The rising band is unusually tight because this historical closed-trade sample contains very limited realized losses. P5, median and P95 full-horizon net profits are USD 523.74, USD 565.35 and USD 609.46. All 1,000 resampled paths ended profitable, with no observed USD 700 floor breach.

The P95 closed-trade drawdown is only 0.10%, compared with 24.86% historical equity drawdown. This large gap is a methodological warning, not evidence that the historical equity excursion can be ignored. Resampling cannot introduce an adverse market regime or an unobserved large basket loss; zero sampled breaches is not a guarantee against future account loss.

MC profit: P5 $523.74 · median $565.35 · P95 $609.46. Profitable 100.0% · USD 700 floor breach 0.0%.

MC statistics and plotted band: matched original cTrader interface JSON export for this run.

A pointwise band does not give a 90% chance that an entire future path stays inside it. Floor breaches are finite-model frequencies, not account-ruin probabilities. Closed-trade resampling omits intra-basket floating losses and broker stop-out.

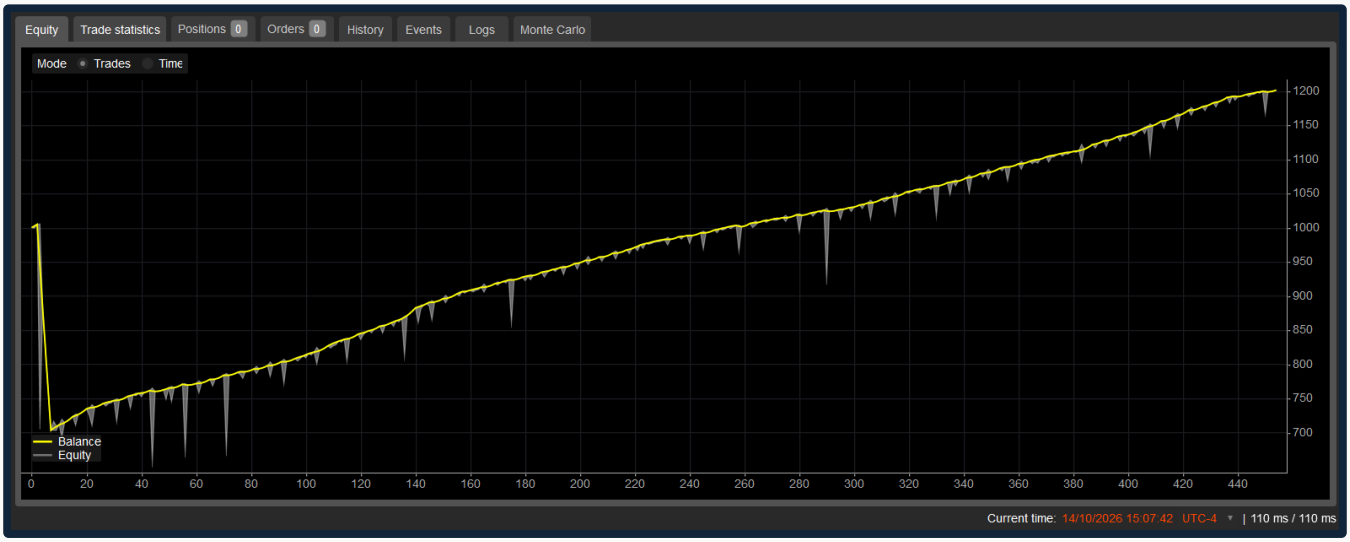
PAIR PROFILE

AUDUSD

H1 · 1 January 2020–14 October 2026 · Historical simulation

NET PROFIT	CUMULATIVE RETURN	MAX. EQUITY DD
+$201.46	20.15%	34.70%

HISTORICAL BALANCE AND EQUITY



cTrader capture: AUDUSD.PNG. Yellow = balance; grey = floating equity. Axis = closed trades; the dated discussion uses native timestamps.

The early downward step is a realized loss: five buy legs closed for −USD 301.57 on 18 March 2020 during the COVID-era FX disruption. The later upward staircase represents gradual rebuilding. The largest cumulative equity drawdown, 34.70%, occurred on 1 September 2020, not in March: the earlier realized loss was then combined with a later sell basket's floating loss.

At that trough equity was USD 656.45 and balance USD 761.32. The sell basket later closed at +USD 0.98 on 22 September. The old March equity peak was not first touched again until 14 June 2024. Final equity reached USD 1,201.46. A positive final return therefore masks a substantial initial loss and a recovery measured in years.

Basket outcome and time at risk	Archived measurement
All closed baskets / mean net	153 / $1.32
Median / P95 / longest holding time	0.18 / 6.31 / 29.56 calendar days
Time in market / baskets reaching five legs	7.91% / 32.68%

Initial $1,000.00 · final equity $1,201.46 · 454 closed legs · leg win rate 92.29% · leg PF 1.65.

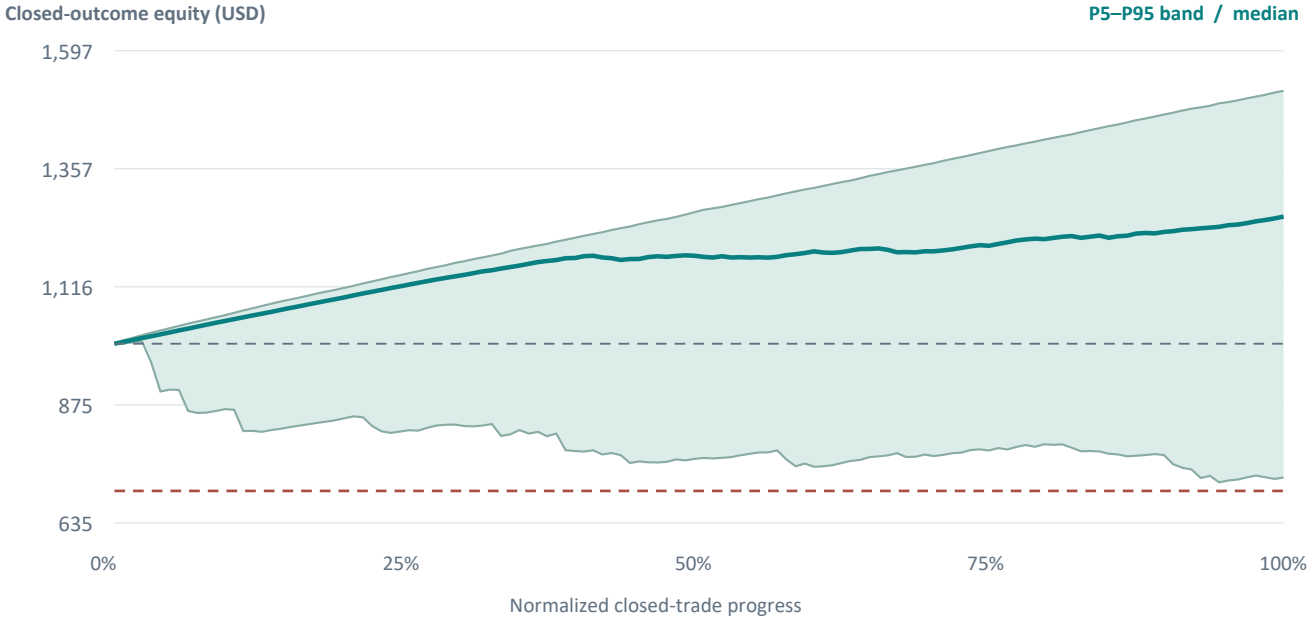
All-closed basket measurements above include shutdowns. One isolated historical test; run ID: 20adab979d1d. Calendar exposure is not a margin or leverage measurement.

PAIR ANALYSIS / MONTE CARLO

AUDUSD · resampled outcomes

Matched closed-trade BlockBootstrap · 1,000 iterations · Five-trade blocks · Seed 0

P5 NET PROFIT	MEDIAN NET PROFIT	P95 NET PROFIT
−$272.83	$259.09	$515.61



Teal line = median; shaded band = pointwise P5–P95 equity; grey dashed line = USD 1,000 start; red dashed line = USD 700 floor. The 128 plotted samples are interpolated over trade progress, not future calendar dates.

The envelope spreads widely because the closed-trade sample includes a large realized loss as well as many smaller gains. Full-horizon net profit is −USD 272.83 at P5, USD 259.09 at the median and USD 515.61 at P95. Profitable endings occur in 82.8% of paths; 9.5% breach the USD 700 floor at some point.

P5 final equity is USD 727.17, above the floor, yet the breach frequency is 9.5% because a path can cross the floor and later recover. That distinction separates terminal outcomes from path risk. The 44.65% P95 closed-trade drawdown describes the bootstrap model, with no new price path, intra-basket floating loss or broker stop-out simulation.

MC profit: P5 −$272.83 · median $259.09 · P95 $515.61. Profitable 82.8% · USD 700 floor breach 9.5%.

MC statistics and plotted band: matched original cTrader interface JSON export for this run.

A pointwise band does not give a 90% chance that an entire future path stays inside it. Floor breaches are finite-model frequencies, not account-ruin probabilities. Closed-trade resampling omits intra-basket floating losses and broker stop-out.

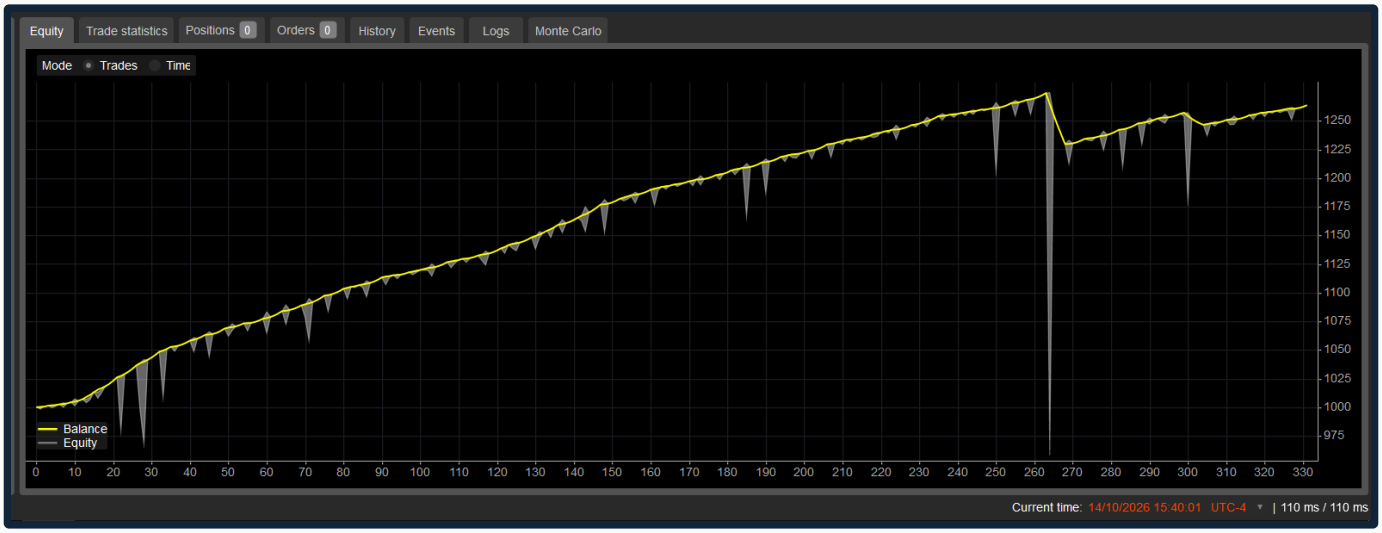
PAIR PROFILE

USDCAD

H1 · 1 January 2020–14 October 2026 · Historical simulation

NET PROFIT	CUMULATIVE RETURN	MAX. EQUITY DD
+$263.41	26.34%	24.46%

HISTORICAL BALANCE AND EQUITY



cTrader capture: USDCAD.PNG. Yellow = balance; grey = floating equity. Axis = closed trades; the dated discussion uses native timestamps.

The balance climbs for much of the test, then shows realized setbacks in 2025. The largest equity decline spans the sampled peak of USD 1,274.44 on 26 November 2024 and the USD 962.65 trough on 3 February 2025: 24.46%. A five-leg sell basket was open through this episode. Canadian-dollar weakness and tariff uncertainty provide context, rather than proof of a sole cause; see pages 41–42.

The basket subsequently closed at −USD 44.49 on 10 April, much smaller than its floating excursion of approximately USD 311.34. The old equity peak was not regained by the end of the run, although final equity remained profitable at USD 1,263.41. Its 86 critical log messages also require operational review.

Basket outcome and time at risk	Archived measurement
All closed baskets / mean net	113 / $2.33
Median / P95 / longest holding time	0.11 / 4.87 / 135.92 calendar days
Time in market / baskets reaching five legs	9.46% / 29.20%

Initial $1,000.00 · final equity $1,263.41 · 331 closed legs · leg win rate 96.07% · leg PF 5.78.

Evidence note: 86 critical log messages (EntityNotFound).

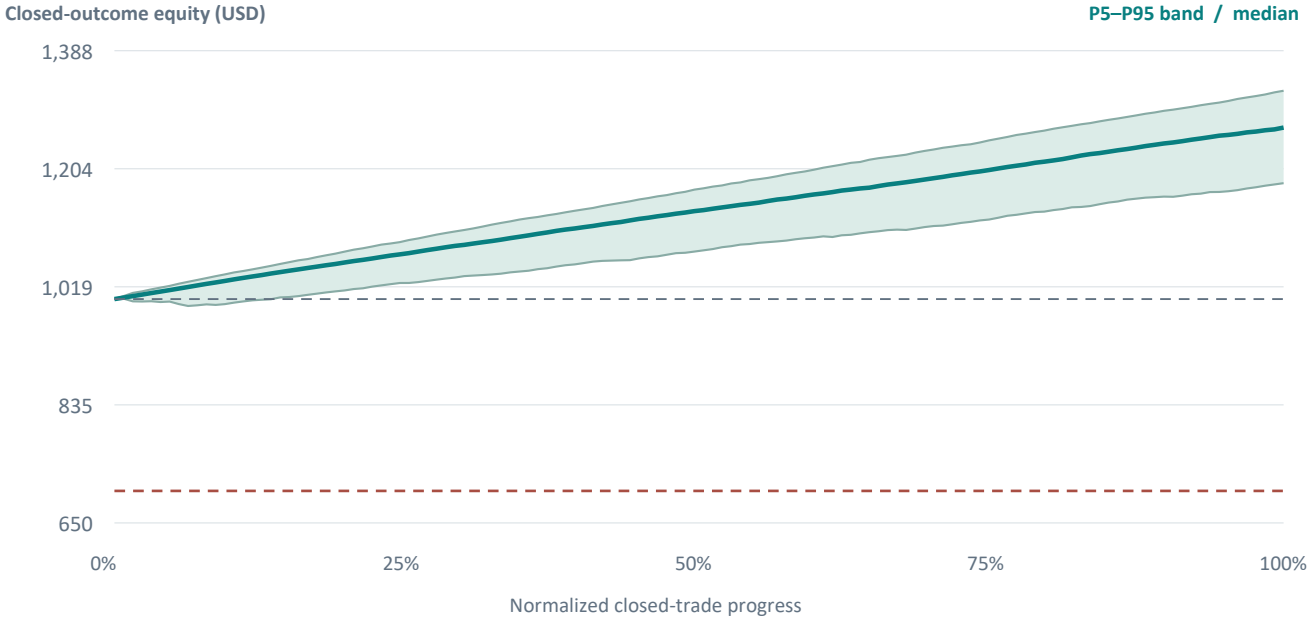
All-closed basket measurements above include shutdowns. One isolated historical test; run ID: cf84560ea8e6. Calendar exposure is not a margin or leverage measurement.

PAIR ANALYSIS / MONTE CARLO

USDCAD · resampled outcomes

Matched closed-trade BlockBootstrap · 1,000 iterations · Five-trade blocks · Seed 0

P5 NET PROFIT	MEDIAN NET PROFIT	P95 NET PROFIT
$181.26	$268.21	$325.80



Teal line = median; shaded band = pointwise P5–P95 equity; grey dashed line = USD 1,000 start; red dashed line = USD 700 floor. The 128 plotted samples are interpolated over trade progress, not future calendar dates.

The median rises while the widening envelope reflects the possibility of drawing the recorded realized losses different numbers of times. P5, median and P95 full-horizon net profits are USD 181.26, USD 268.21 and USD 325.80. All 1,000 sampled paths ended profitable and none breached the USD 700 floor.

The 4.78% P95 resampled closed-trade drawdown should be compared with the historical 24.46% equity drawdown. The bootstrap sees the eventual recorded loss, not the larger floating decline that preceded it. It also does not replay the execution conditions associated with the 86 critical log messages. These limits matter when judging suitability or increasing position size.

MC profit: P5 $181.26 · median $268.21 · P95 $325.80. Profitable 100.0% · USD 700 floor breach 0.0%.

MC statistics and plotted band: matched original cTrader interface JSON export for this run.

A pointwise band does not give a 90% chance that an entire future path stays inside it. Floor breaches are finite-model frequencies, not account-ruin probabilities. Closed-trade resampling omits intra-basket floating losses and broker stop-out.

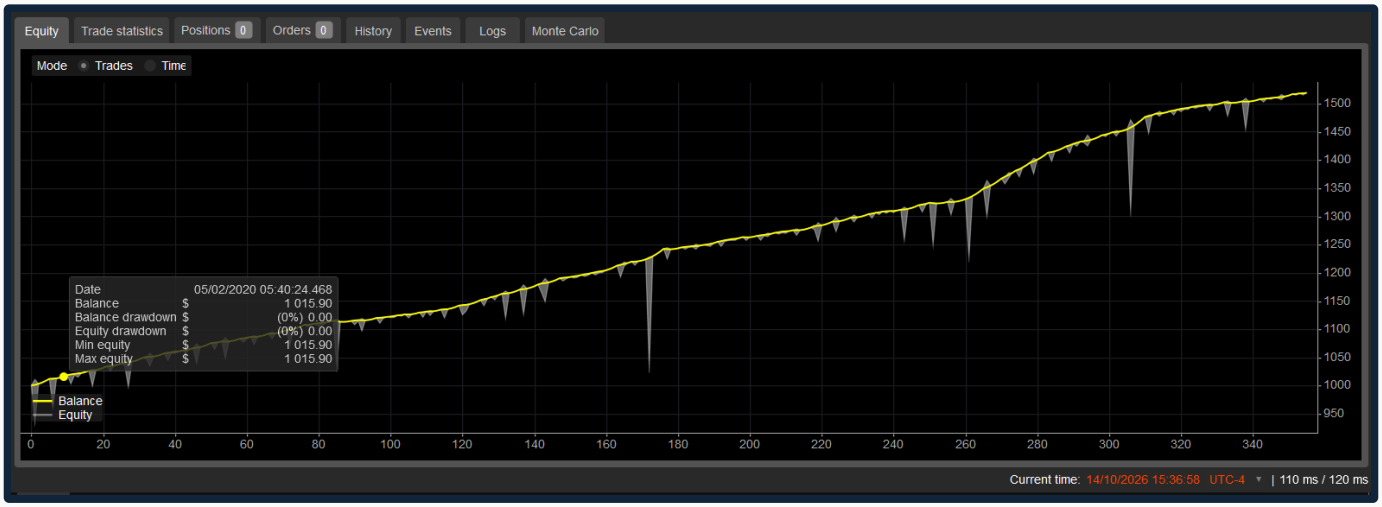
PAIR PROFILE

USDCHF

H1 · 1 January 2020–14 October 2026 · Historical simulation

NET PROFIT	CUMULATIVE RETURN	MAX. EQUITY DD
+$518.93	51.89%	15.75%

HISTORICAL BALANCE AND EQUITY



cTrader capture: USDCHF.PNG. Yellow = balance; grey = floating equity. Axis = closed trades; the dated discussion uses native timestamps.

The yellow balance line rises to USD 1,518.93, but the grey path shows recurring open-basket losses. The largest sampled equity decline was 15.75%, from USD 1,224.69 on 19 December 2023 to USD 1,031.84 on 28 December 2023. A five-leg buy basket was exposed at the trough; the records establish the adverse position direction without isolating one macroeconomic cause.

The basket closed at +USD 18.68 on 17 January 2024, the same date on which equity first touched the old peak in the sampled record. Balance drawdown was only 0.22%. The profitable exit should not disguise the preceding floating loss. This historically strong ROI result also contains 39 critical messages, which need reconciliation before deployment.

Basket outcome and time at risk	Archived measurement
All closed baskets / mean net	125 / $4.15
Median / P95 / longest holding time	0.15 / 5.05 / 36.79 calendar days
Time in market / baskets reaching five legs	6.93% / 28.80%

Initial $1,000.00 · final equity $1,518.93 · 355 closed legs · leg win rate 94.37% · leg PF 59.97.

Evidence note: 39 critical log messages (EntityNotFound).

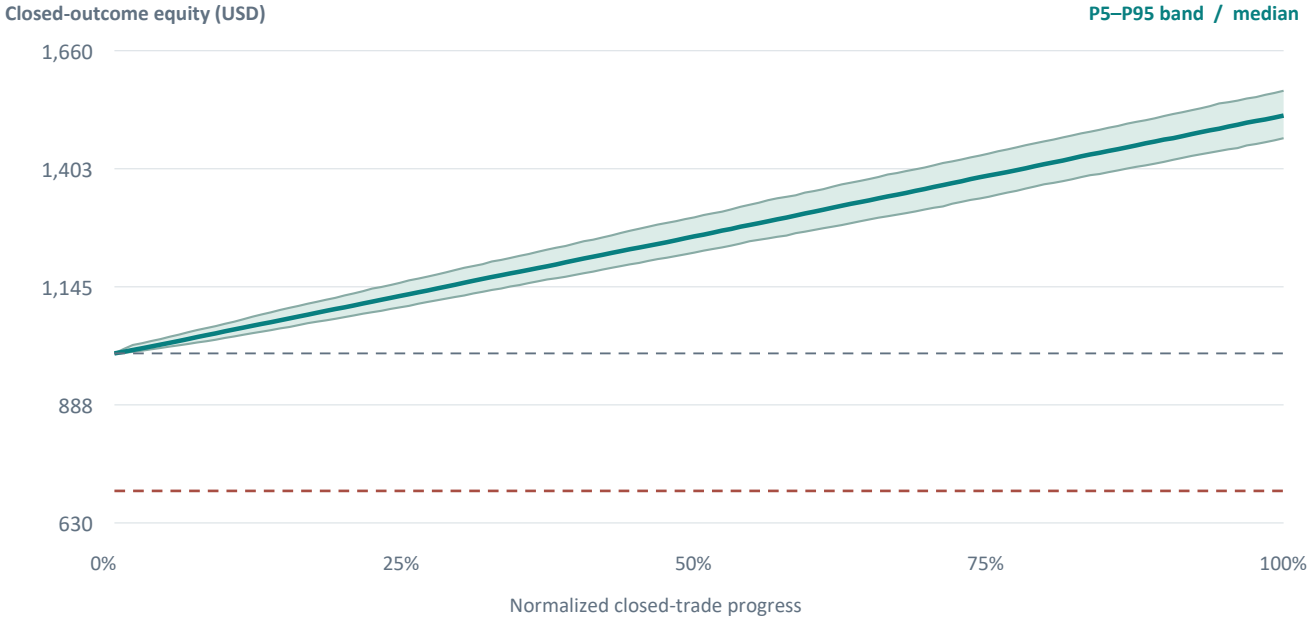
All-closed basket measurements above include shutdowns. One isolated historical test; run ID: 9f26b493dc69. Calendar exposure is not a margin or leverage measurement.

PAIR ANALYSIS / MONTE CARLO

USDCHF · resampled outcomes

Matched closed-trade BlockBootstrap · 1,000 iterations · Five-trade blocks · Seed 0

P5 NET PROFIT	MEDIAN NET PROFIT	P95 NET PROFIT
$469.47	$518.81	$572.97



Teal line = median; shaded band = pointwise P5–P95 equity; grey dashed line = USD 1,000 start; red dashed line = USD 700 floor. The 128 plotted samples are interpolated over trade progress, not future calendar dates.

The tight upward band reflects mostly positive recorded closed outcomes. P5, median and P95 full-horizon net profits are USD 469.47, USD 518.81 and USD 572.97. All 1,000 resampled paths ended profitable, with no observed breach of the USD 700 floor.

The P95 closed-trade drawdown is just 0.23%, compared with 15.75% historical equity drawdown. This difference illustrates what the bootstrap omits: a basket can endure a substantial open loss and then contribute only a small positive closing result to the resampled series. The calculation cannot validate execution reliability or remove the need to investigate the 39 critical messages.

MC profit: P5 $469.47 · median $518.81 · P95 $572.97. Profitable 100.0% · USD 700 floor breach 0.0%.

MC statistics and plotted band: matched original cTrader interface JSON export for this run.

A pointwise band does not give a 90% chance that an entire future path stays inside it. Floor breaches are finite-model frequencies, not account-ruin probabilities. Closed-trade resampling omits intra-basket floating losses and broker stop-out.

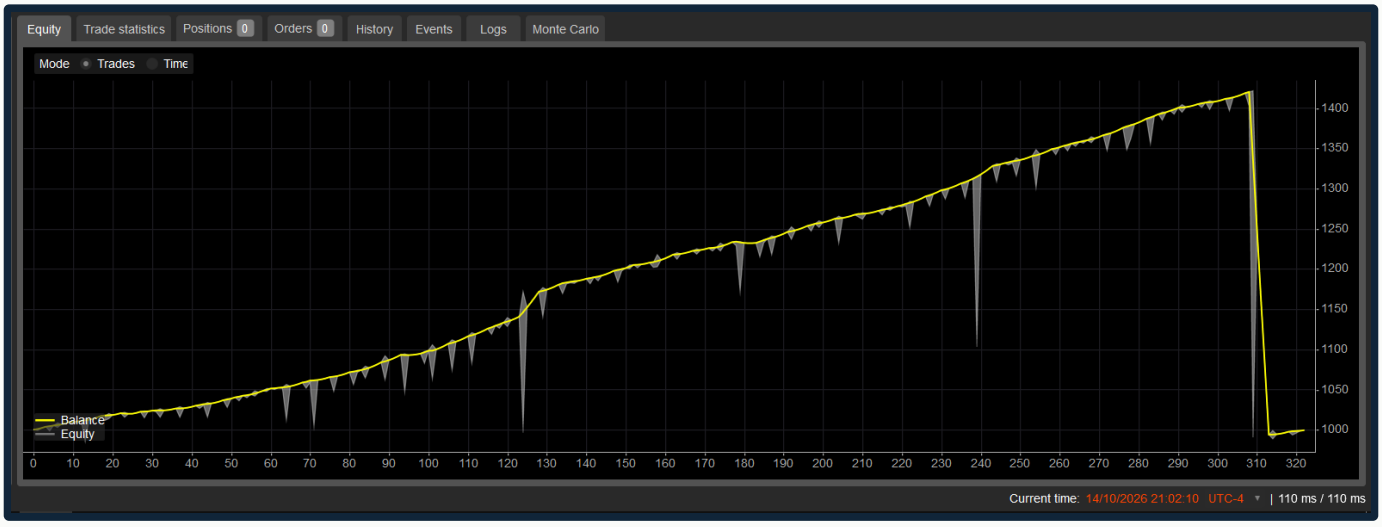
PAIR PROFILE

USDJPY

H1 · 1 January 2020–14 October 2026 · Historical simulation

NET PROFIT	CUMULATIVE RETURN	MAX. EQUITY DD
−$0.61	-0.06%	30.44%

HISTORICAL BALANCE AND EQUITY



cTrader capture: USDJPY.PNG. Yellow = balance; grey = floating equity. Axis = closed trades; the dated discussion uses native timestamps.

The long upward staircase is interrupted by a large late balance collapse: a five-leg sell basket realized −USD 425.97 on 17 June 2026. This single episode erased much of the earlier accumulation. Maximum equity drawdown was 30.44%, measured from the USD 1,421.22 sampled peak on 21 October 2025 to USD 988.56 on 6 July 2026.

The July trough includes the June realized loss plus a later five-leg sell basket's smaller floating loss. That later basket closed at +USD 3.56 on 6 July; it did not recover the account's earlier high. Final equity was USD 999.39, a net −USD 0.61. The 128 critical messages strengthen the case for execution review; no sole macroeconomic cause is established.

Basket outcome and time at risk	Archived measurement
All closed baskets / mean net	102 / −$0.01
Median / P95 / longest holding time	0.14 / 8.45 / 239.47 calendar days
Time in market / baskets reaching five legs	14.60% / 36.27%

Initial $1,000.00 · final equity $999.39 · 322 closed legs · leg win rate 95.65% · leg PF 1.00.

Evidence note: 128 critical log messages (EntityNotFound).

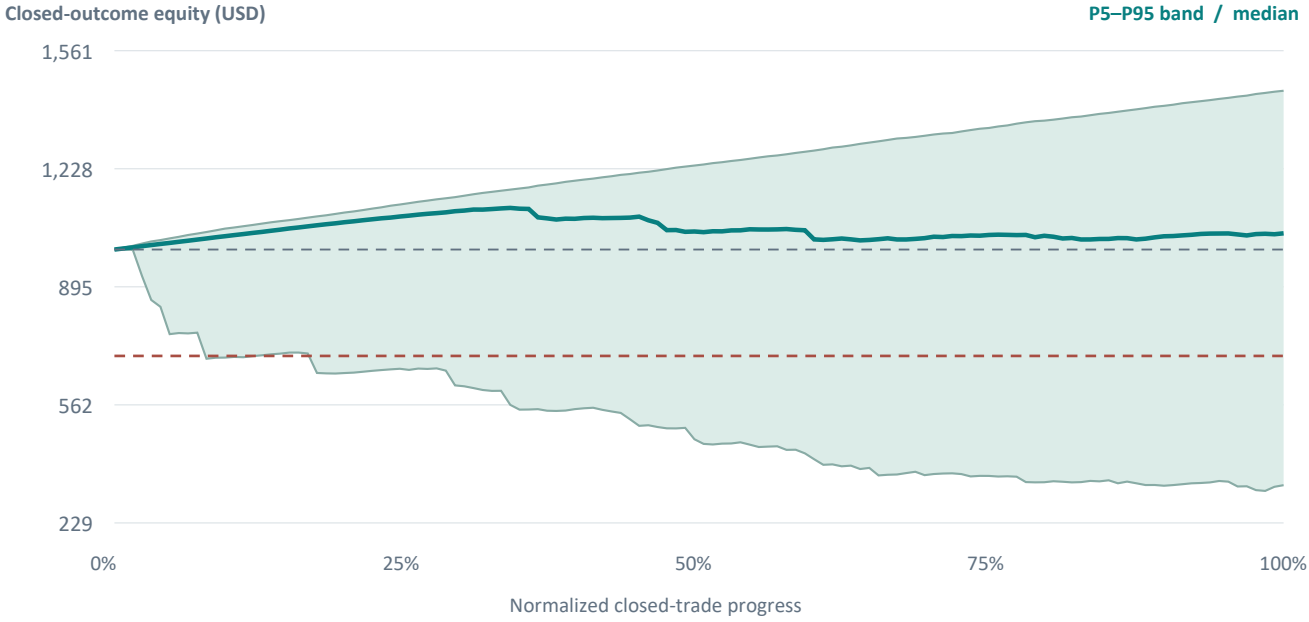
All-closed basket measurements above include shutdowns. One isolated historical test; run ID: 482bc6907f5b. Calendar exposure is not a margin or leverage measurement.

PAIR ANALYSIS / MONTE CARLO

USDJPY · resampled outcomes

Matched closed-trade BlockBootstrap · 1,000 iterations · Five-trade blocks · Seed 0

P5 NET PROFIT	MEDIAN NET PROFIT	P95 NET PROFIT
−$664.39	$45.74	$447.74



Teal line = median; shaded band = pointwise P5–P95 equity; grey dashed line = USD 1,000 start; red dashed line = USD 700 floor. The 128 plotted samples are interpolated over trade progress, not future calendar dates.

The envelope fans out sharply because resampling can omit, retain or repeat the recorded large losing sequence. P5 full-horizon net profit is −USD 664.39, the median USD 45.74 and P95 USD 447.74. Only 55.4% of paths end profitable; 32.8% breach the USD 700 floor. A modestly positive median therefore coexists with a large modeled downside.

The 79.49% P95 closed-trade drawdown is model output, not a forecast or a broker-aware liquidation path. The supplied matched calculation uses the local cTrader engine. Its 1,000 paths do not model intra-basket equity, margin intervention or corrections to the archived 128 critical log messages.

MC profit: P5 −$664.39 · median $45.74 · P95 $447.74. Profitable 55.4% · USD 700 floor breach 32.8%.

MC statistics: local cTrader engine calculation from this run's closed trades; not an interface JSON export.

A pointwise band does not give a 90% chance that an entire future path stays inside it. Floor breaches are finite-model frequencies, not account-ruin probabilities. Closed-trade resampling omits intra-basket floating losses and broker stop-out.

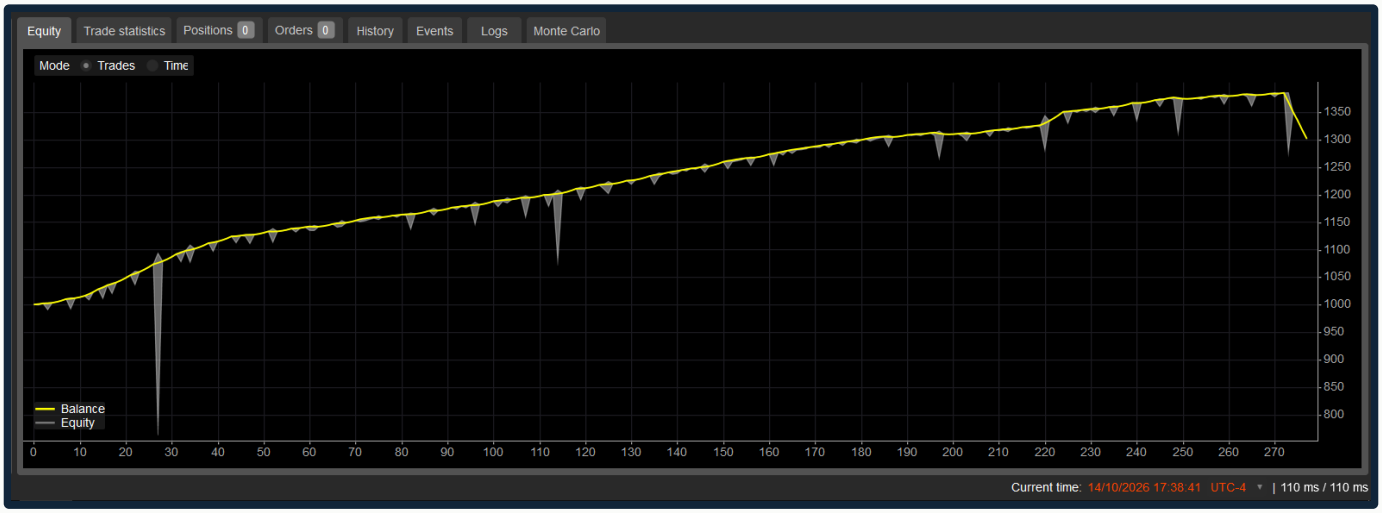
PAIR PROFILE

AUDCHF

H1 · 1 January 2020–14 October 2026 · Historical simulation

NET PROFIT	CUMULATIVE RETURN	MAX. EQUITY DD
+$301.42	30.14%	28.23%

HISTORICAL BALANCE AND EQUITY



cTrader capture: AUDCHF.PNG. Yellow = balance; grey = floating equity. Axis = closed trades; the dated discussion uses native timestamps.

The March 2020 grey plunge is principally a floating-equity event, rather than a matching collapse in realized balance. Equity fell from a sampled USD 1,075.74 peak on 16 March to USD 772.09 on 19 March 2020, a 28.23% drawdown, with five buy legs exposed during COVID-era FX stress.

The old peak was first touched again in the sampled record on 20 March, while the basket itself closed at +USD 18.44 on 25 March. A temporary equity recovery and a completed profitable exit are different observations. After the later accumulation, a final sell basket was liquidated at shutdown for −USD 83.30 on 14 October 2026. That terminal step is included in final equity of USD 1,301.42.

Basket outcome and time at risk	Archived measurement
All closed baskets / mean net	104 / $2.90
Median / P95 / longest holding time	0.18 / 6.68 / 54.42 calendar days
Time in market / baskets reaching five legs	6.47% / 27.88%

Initial $1,000.00 · final equity $1,301.42 · 277 closed legs · leg win rate 90.25% · leg PF 4.18.

Evidence note: 6 critical log messages (EntityNotFound); one end-of-run basket excluded from completed-basket statistics.

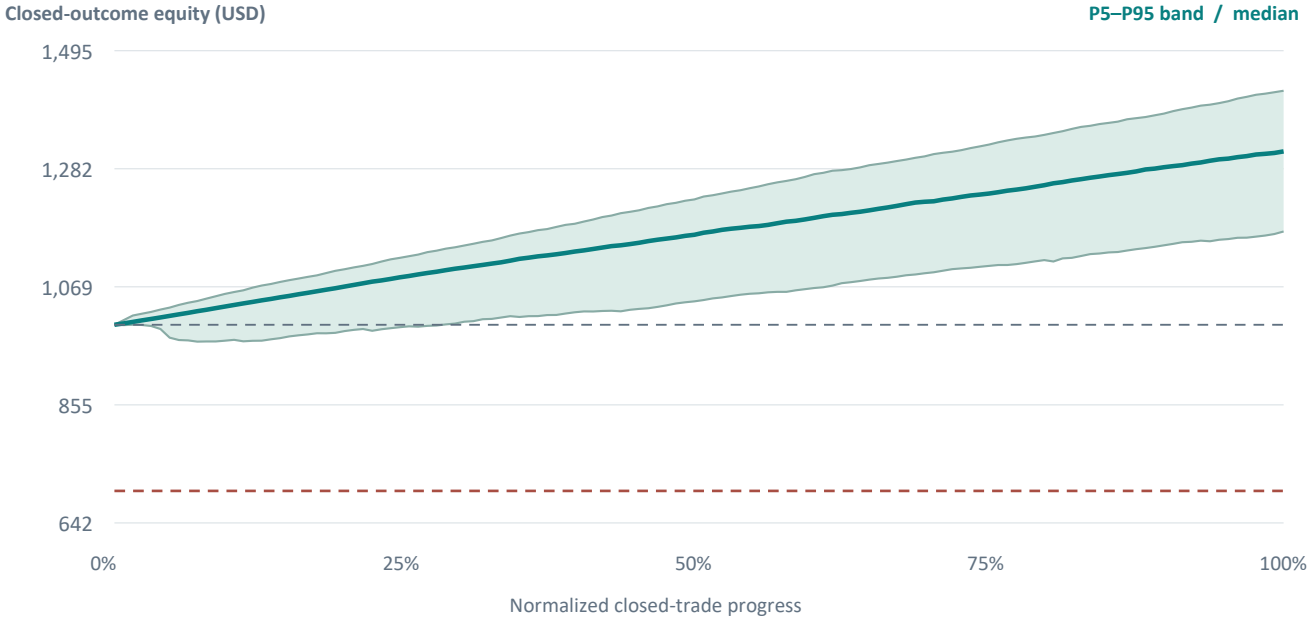
All-closed basket measurements above include shutdowns. One isolated historical test; run ID: 30c6e40f88e8. Calendar exposure is not a margin or leverage measurement.

PAIR ANALYSIS / MONTE CARLO

AUDCHF · resampled outcomes

Matched closed-trade BlockBootstrap · 1,000 iterations · Five-trade blocks · Seed 0

P5 NET PROFIT	MEDIAN NET PROFIT	P95 NET PROFIT
$168.47	$313.09	$422.62



Teal line = median; shaded band = pointwise P5–P95 equity; grey dashed line = USD 1,000 start; red dashed line = USD 700 floor. The 128 plotted samples are interpolated over trade progress, not future calendar dates.

The rising but widening envelope reflects mainly profitable closed outcomes together with the recorded realized losses, including the shutdown basket. Full-horizon P5, median and P95 net profits are USD 168.47, USD 313.09 and USD 422.62. Profitable endings occur in 99.5% of paths; no USD 700 floor breach was observed in 1,000 iterations.

The 9.53% P95 closed-trade drawdown is distinct from the 28.23% historical equity drawdown. The bootstrap includes the final recorded loss, but it cannot reconstruct the earlier COVID-era floating excursion from the basket's eventual positive close. It also rearranges a forced test-end liquidation, which should not be mistaken for a naturally repeated trading exit.

MC profit: P5 $168.47 · median $313.09 · P95 $422.62. Profitable 99.5% · USD 700 floor breach 0.0%.

MC statistics and plotted band: matched original cTrader interface JSON export for this run.

A pointwise band does not give a 90% chance that an entire future path stays inside it. Floor breaches are finite-model frequencies, not account-ruin probabilities. Closed-trade resampling omits intra-basket floating losses and broker stop-out.

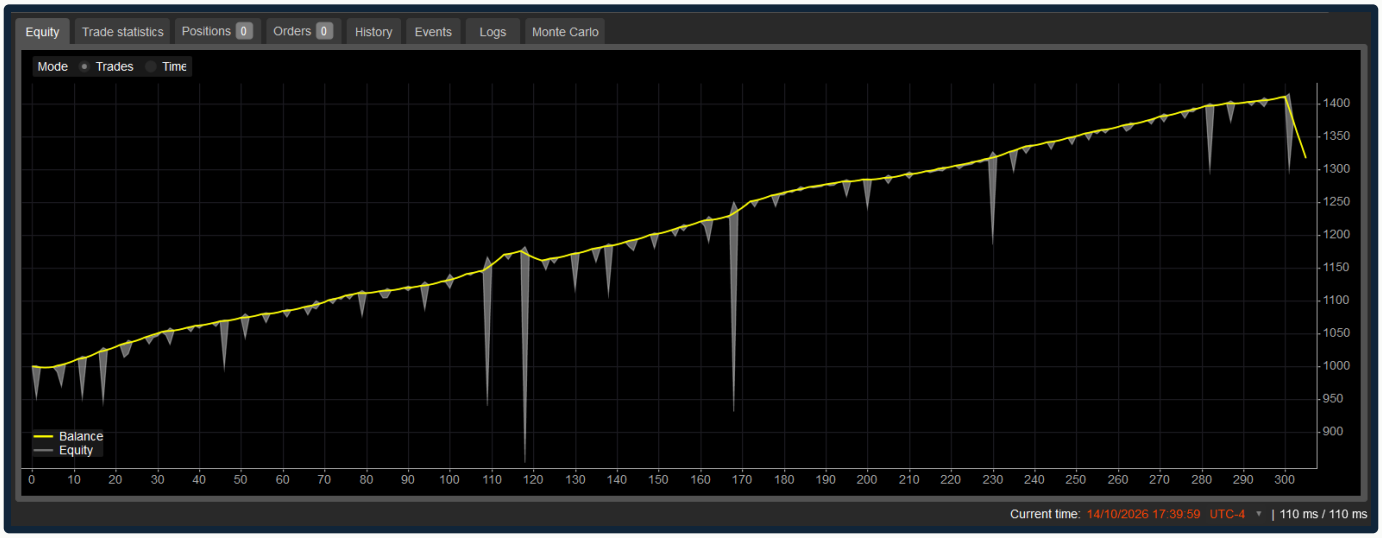
PAIR PROFILE

EURAUD

H1 · 1 January 2020–14 October 2026 · Historical simulation

NET PROFIT	CUMULATIVE RETURN	MAX. EQUITY DD
+$316.63	31.66%	27.11%

HISTORICAL BALANCE AND EQUITY



cTrader capture: EURAUD.PNG. Yellow = balance; grey = floating equity. Axis = closed trades; the dated discussion uses native timestamps.

The broadly rising balance path contains equity dips that are much larger than the eventual realized setbacks. The maximum equity drawdown was 27.11%: from a sampled USD 1,181.94 peak on 16 March 2022 to USD 861.57 on 5 April 2022, while five buy legs were open. This basket later closed at −USD 14.58 on 10 May.

The account first touched the old equity peak in a sampled interval on 19 August 2022. A later terminal decline reflects a separate buy basket liquidated at shutdown for −USD 93.33 on 14 October 2026; it is not the 2022 drawdown event. Final equity was USD 1,316.63. Nine critical messages remain part of the execution evidence, and the records do not establish a sole macroeconomic cause.

Basket outcome and time at risk	Archived measurement
All closed baskets / mean net	101 / $3.13
Median / P95 / longest holding time	0.09 / 11.10 / 84.03 calendar days
Time in market / baskets reaching five legs	13.40% / 38.61%

Initial $1,000.00 · final equity $1,316.63 · 305 closed legs · leg win rate 94.43% · leg PF 3.86.

Evidence note: 9 critical log messages (EntityNotFound); one end-of-run basket excluded from completed-basket statistics.

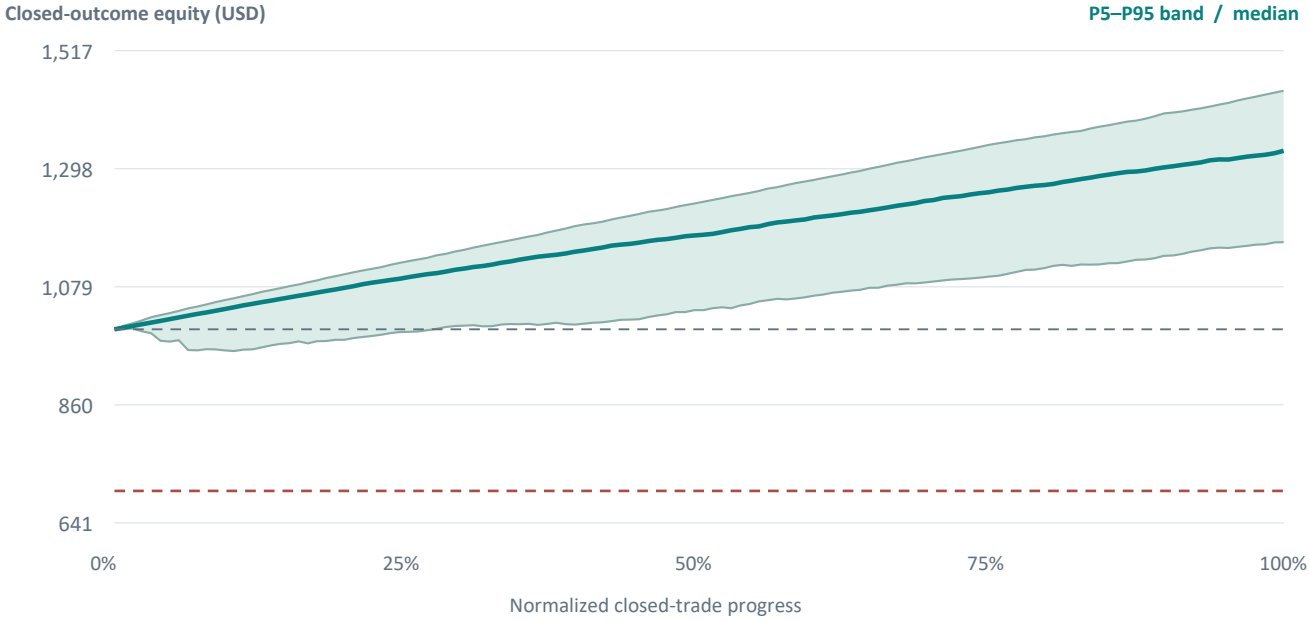
All-closed basket measurements above include shutdowns. One isolated historical test; run ID: 549c6dd44212. Calendar exposure is not a margin or leverage measurement.

PAIR ANALYSIS / MONTE CARLO

EURAUD · resampled outcomes

Matched closed-trade BlockBootstrap · 1,000 iterations · Five-trade blocks · Seed 0

P5 NET PROFIT	MEDIAN NET PROFIT	P95 NET PROFIT
$161.86	$331.34	$442.87



Teal line = median; shaded band = pointwise P5–P95 equity; grey dashed line = USD 1,000 start; red dashed line = USD 700 floor. The 128 plotted samples are interpolated over trade progress, not future calendar dates.

The median rises, while the broader lower envelope reflects the recorded losing baskets and their possible repetition in resampled sequences. P5, median and P95 full-horizon net profits are USD 161.86, USD 331.34 and USD 442.87. Profitable endings occur in 99.9% of paths, with no observed breach of the USD 700 floor.

The 11.13% P95 closed-trade drawdown does not capture the historical 27.11% floating-equity excursion. The final shutdown basket's loss is present among the closed outcomes; resampling it does not make shutdown a recurring strategy rule. Neither the high profitable frequency nor zero observed floor breaches demonstrates that a larger position size would preserve these results.

MC profit: P5 $161.86 · median $331.34 · P95 $442.87. Profitable 99.9% · USD 700 floor breach 0.0%.

MC statistics and plotted band: matched original cTrader interface JSON export for this run.

A pointwise band does not give a 90% chance that an entire future path stays inside it. Floor breaches are finite-model frequencies, not account-ruin probabilities. Closed-trade resampling omits intra-basket floating losses and broker stop-out.

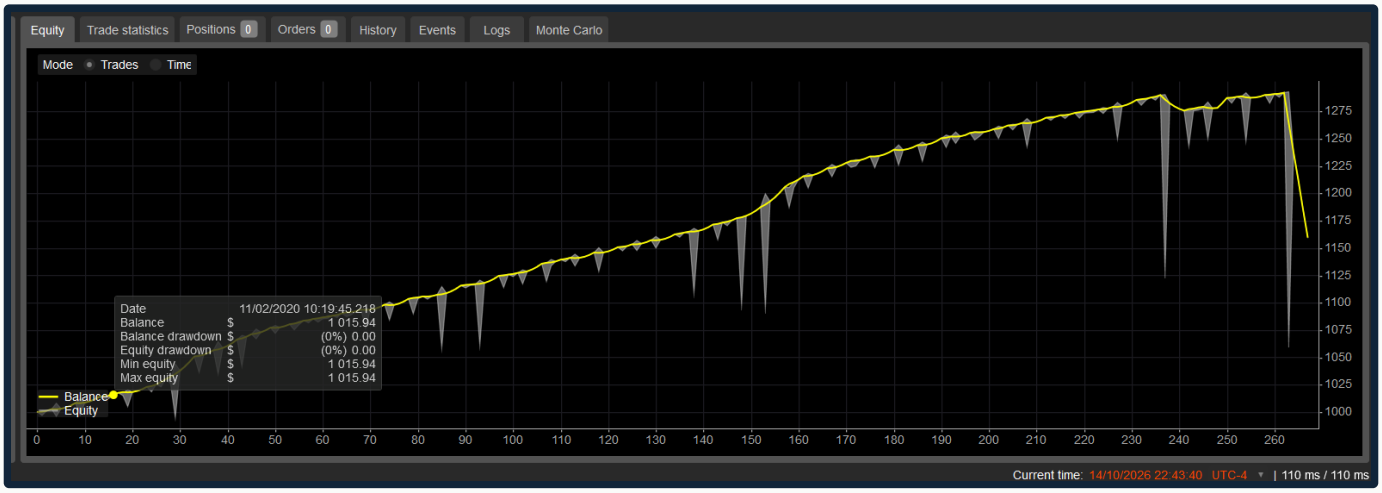
PAIR PROFILE

EURCHF

H1 · 1 January 2020–14 October 2026 · Historical simulation

NET PROFIT	CUMULATIVE RETURN	MAX. EQUITY DD
+$159.35	15.93%	17.70%

HISTORICAL BALANCE AND EQUITY



cTrader capture: EURCHF.PNG. Yellow = balance; grey = floating equity. Axis = closed trades; the dated discussion uses native timestamps.

The balance rises for much of the study, while the final grey decline develops inside a basket that remains open until the test ends. The largest equity drawdown was 17.70%, from a sampled USD 1,292.64 peak on 6 July 2026 to USD 1,063.83 on 30 September 2026. Five sell legs were exposed at that trough.

That basket closed at −USD 132.49 during shutdown on 14 October 2026, creating the terminal realized-balance step. The records classify it as EndOfRun, not an established stop-out. The old equity peak was not regained before the run ended. Final equity was USD 1,159.35. Its final loss is included in total profit, although completed-basket statistics exclude this shutdown basket.

Basket outcome and time at risk	Archived measurement
All closed baskets / mean net	92 / $1.73
Median / P95 / longest holding time	0.31 / 10.40 / 103.21 calendar days
Time in market / baskets reaching five legs	9.60% / 28.26%

Initial $1,000.00 · final equity $1,159.35 · 267 closed legs · leg win rate 88.39% · leg PF 2.04.

Evidence note: 4 critical log messages (EntityNotFound); one end-of-run basket excluded from completed-basket statistics.

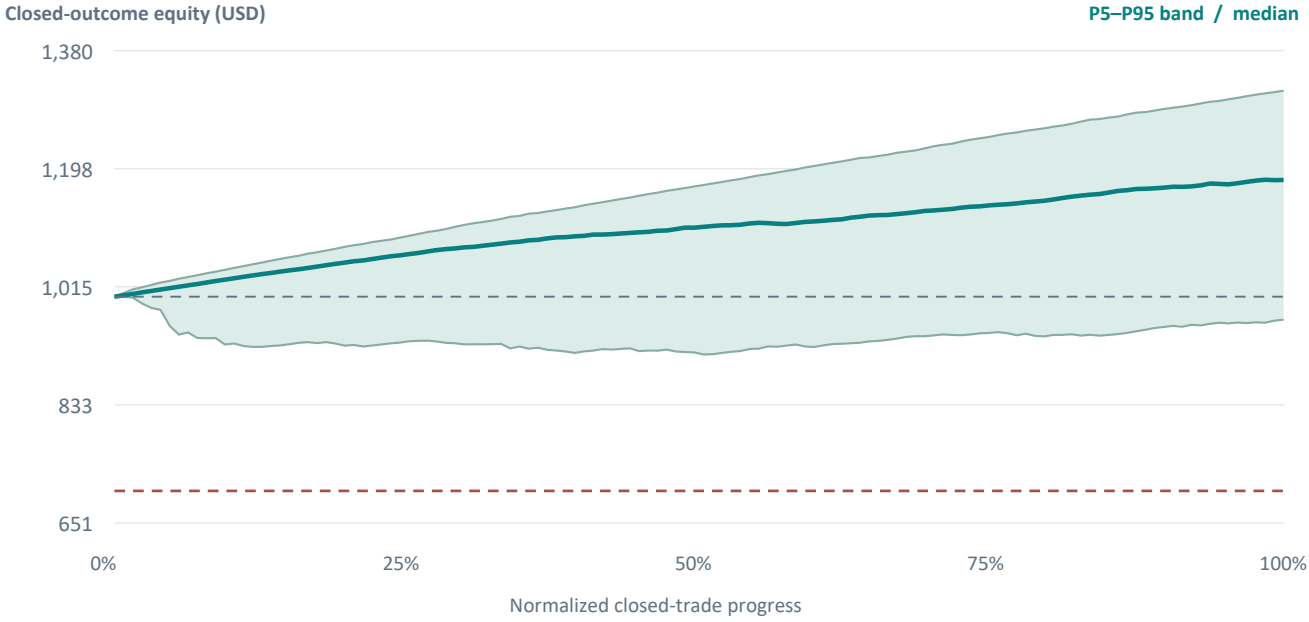
All-closed basket measurements above include shutdowns. One isolated historical test; run ID: e8166f735361. Calendar exposure is not a margin or leverage measurement.

PAIR ANALYSIS / MONTE CARLO

EURCHF · resampled outcomes

Matched closed-trade BlockBootstrap · 1,000 iterations · Five-trade blocks · Seed 0

P5 NET PROFIT	MEDIAN NET PROFIT	P95 NET PROFIT
−$35.45	$180.22	$318.06



Teal line = median; shaded band = pointwise P5–P95 equity; grey dashed line = USD 1,000 start; red dashed line = USD 700 floor. The 128 plotted samples are interpolated over trade progress, not future calendar dates.

The widening band crosses the original USD 1,000 level at its lower edge. Full-horizon net profit is −USD 35.45 at P5, USD 180.22 at the median and USD 318.06 at P95. Profitable endings occur in 91.0% of paths; 0.8% breach the USD 700 floor. A positive median therefore does not imply uniformly positive outcomes.

This matched local-engine calculation includes the final shutdown loss among the observed closed trades. Repeating or omitting that sequence changes the envelope, but does not replay new EURCHF prices. Its 19.60% P95 closed-trade drawdown is a resampled measure, distinct from the historical 17.70% equity drawdown and from a future margin or stop-out probability.

MC profit: P5 −$35.45 · median $180.22 · P95 $318.06. Profitable 91.0% · USD 700 floor breach 0.8%.

MC statistics: local cTrader engine calculation from this run's closed trades; not an interface JSON export.

A pointwise band does not give a 90% chance that an entire future path stays inside it. Floor breaches are finite-model frequencies, not account-ruin probabilities. Closed-trade resampling omits intra-basket floating losses and broker stop-out.

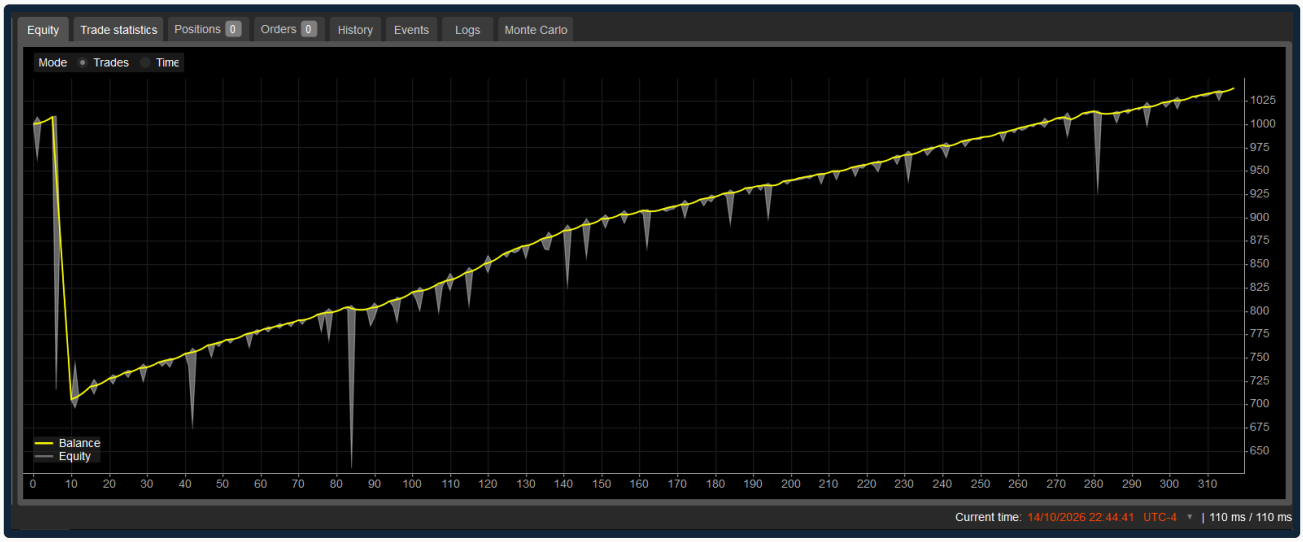
PAIR PROFILE

NZDUSD

H1 · 1 January 2020–14 October 2026 · Historical simulation

NET PROFIT	CUMULATIVE RETURN	MAX. EQUITY DD
+$38.40	3.84%	36.84%

HISTORICAL BALANCE AND EQUITY



cTrader capture: NZDUSD.PNG. Yellow = balance; grey = floating equity. Axis = closed trades; the dated discussion uses native timestamps.

The abrupt early balance decline records a five-buy-leg loss of −USD 302.32 on 19 March 2020 during COVID-era FX stress. Slow subsequent accumulation did not prevent a deeper cumulative equity trough on 28 January 2022. Equity then reached USD 636.97, a 36.84% drawdown from the old USD 1,008.44 March 2020 peak.

The later trough combines the earlier realized loss with approximately USD 166.95 of floating loss in another five-leg buy basket. That basket closed at −USD 2.30 on 6 March 2022; January 2022 should not be labeled a fresh COVID crash. The old equity peak was first touched again on 8 October 2025. Final equity was only USD 1,038.40, illustrating a long recovery and limited net accumulation.

Basket outcome and time at risk	Archived measurement
All closed baskets / mean net	111 / $0.35
Median / P95 / longest holding time	0.18 / 6.71 / 51.73 calendar days
Time in market / baskets reaching five legs	7.13% / 32.43%

Initial $1,000.00 · final equity $1,038.40 · 317 closed legs · leg win rate 91.48% · leg PF 1.12.

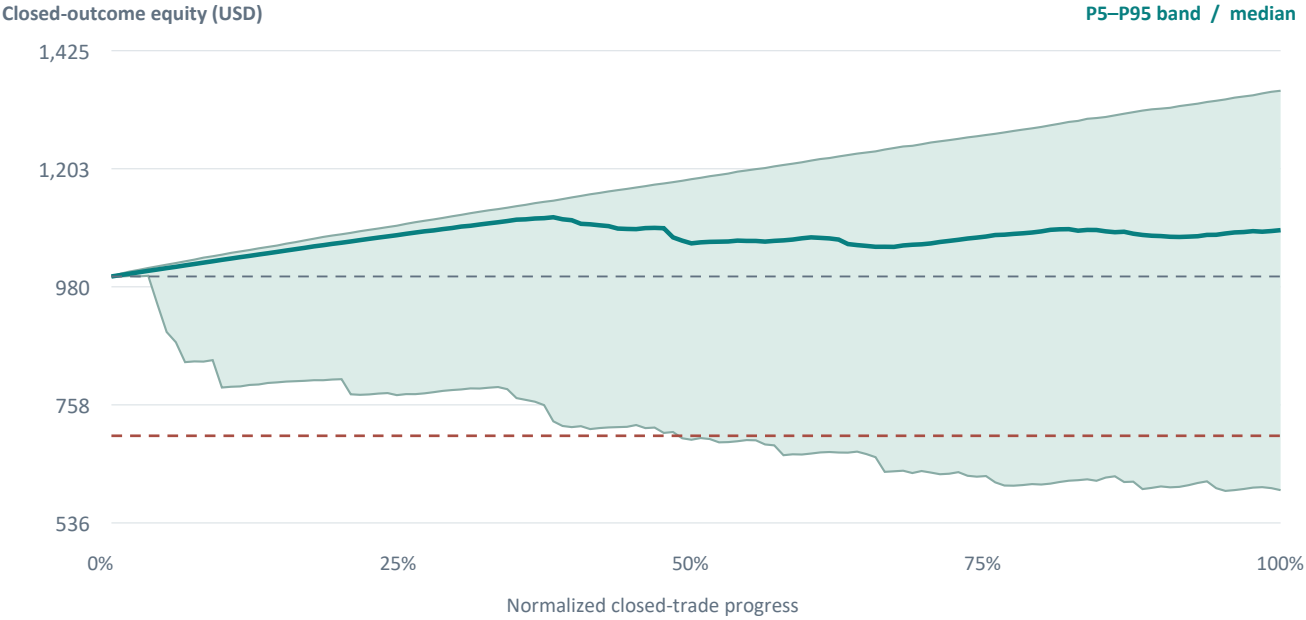
All-closed basket measurements above include shutdowns. One isolated historical test; run ID: ae7da8d6d7de. Calendar exposure is not a margin or leverage measurement.

PAIR ANALYSIS / MONTE CARLO

NZDUSD · resampled outcomes

Matched closed-trade BlockBootstrap · 1,000 iterations · Five-trade blocks · Seed 0

P5 NET PROFIT	MEDIAN NET PROFIT	P95 NET PROFIT
−$402.35	$87.20	$349.68



Teal line = median; shaded band = pointwise P5–P95 equity; grey dashed line = USD 1,000 start; red dashed line = USD 700 floor. The 128 plotted samples are interpolated over trade progress, not future calendar dates.

The broad envelope straddles the original capital level because the large realized 2020 loss can be repeated or omitted by the bootstrap. P5, median and P95 full-horizon net profits are −USD 402.35, USD 87.20 and USD 349.68. Profitable endings occur in 63.1% of paths; 15.1% breach the USD 700 floor.

The positive median should be weighed against that losing lower tail and the historical 36.84% equity drawdown. P95 resampled closed-trade drawdown is 53.19%, but it still excludes the floating trajectory inside each basket and broker margin intervention. The displayed band is a sequence-sensitivity exercise using the observed outcomes, not an estimate of live annual returns.

MC profit: P5 −$402.35 · median $87.20 · P95 $349.68. Profitable 63.1% · USD 700 floor breach 15.1%.

MC statistics and plotted band: matched original cTrader interface JSON export for this run.

A pointwise band does not give a 90% chance that an entire future path stays inside it. Floor breaches are finite-model frequencies, not account-ruin probabilities. Closed-trade resampling omits intra-basket floating losses and broker stop-out.

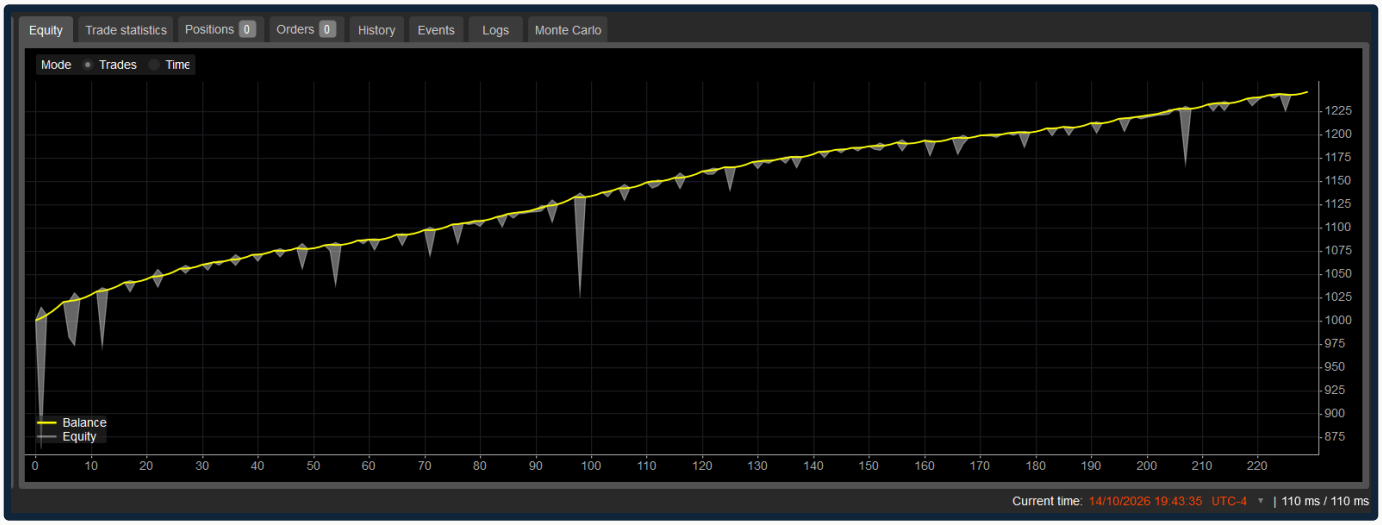
PAIR PROFILE

EURGBP

H1 · 1 January 2020–14 October 2026 · Historical simulation

NET PROFIT	CUMULATIVE RETURN	MAX. EQUITY DD
+$245.75	24.58%	14.46%

HISTORICAL BALANCE AND EQUITY



cTrader capture: EURGBP.PNG. Yellow = balance; grey = floating equity. Axis = closed trades; the dated discussion uses native timestamps.

The yellow balance line remains much steadier than the grey equity path. The largest sampled equity decline occurred on 23 March 2020 during COVID-era FX stress: from USD 1,014.16 to USD 867.55, or 14.46%, with five sell legs exposed. Balance at the trough interval's end was still USD 1,000.

The basket closed at +USD 19.77 on 24 March, when the sampled record first touched the prior equity peak again. That outcome explains why the conspicuous grey drop need not appear as a comparable realized-balance loss. Later gains brought final equity to USD 1,245.75, while balance drawdown was only 0.10%. Five critical messages remain in the logs; a smooth balance line does not establish error-free execution.

Basket outcome and time at risk	Archived measurement
All closed baskets / mean net	81 / $3.03
Median / P95 / longest holding time	0.31 / 5.23 / 27.09 calendar days
Time in market / baskets reaching five legs	4.21% / 29.63%

Initial $1,000.00 · final equity $1,245.75 · 229 closed legs · leg win rate 87.34% · leg PF 23.02.

Evidence note: 5 critical log messages (EntityNotFound).

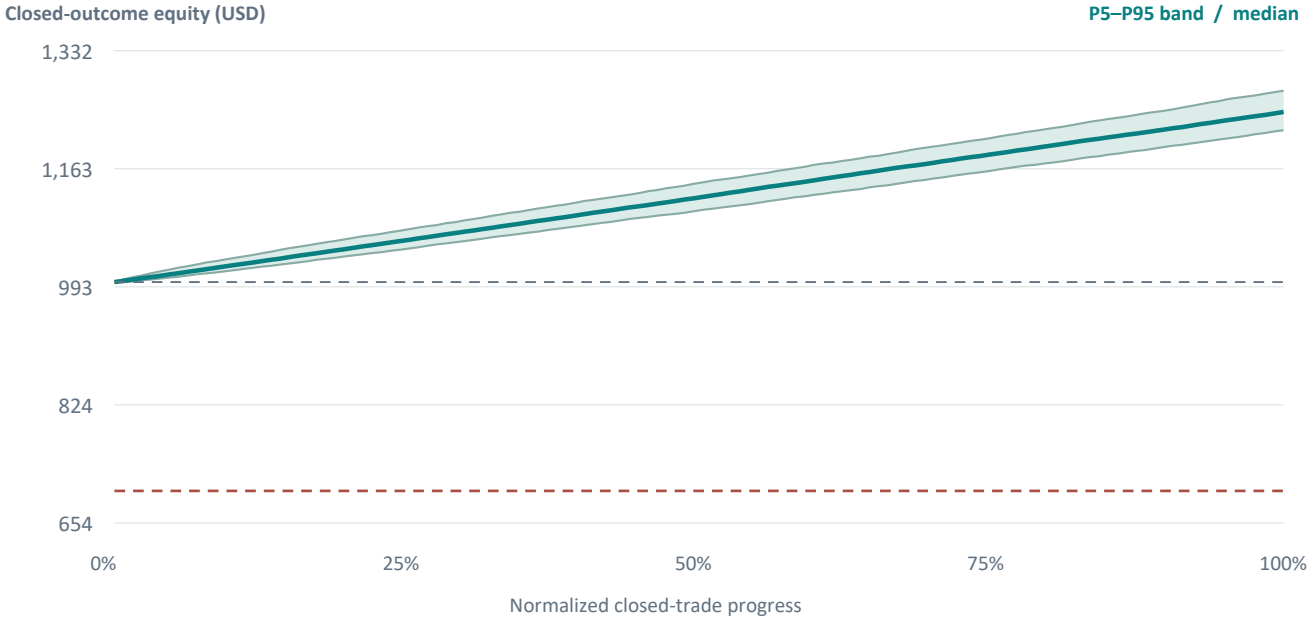
All-closed basket measurements above include shutdowns. One isolated historical test; run ID: 041a08dc289c. Calendar exposure is not a margin or leverage measurement.

PAIR ANALYSIS / MONTE CARLO

EURGBP · resampled outcomes

Matched closed-trade BlockBootstrap · 1,000 iterations · Five-trade blocks · Seed 0

P5 NET PROFIT	MEDIAN NET PROFIT	P95 NET PROFIT
$218.08	$244.43	$274.87



Teal line = median; shaded band = pointwise P5–P95 equity; grey dashed line = USD 1,000 start; red dashed line = USD 700 floor. The 128 plotted samples are interpolated over trade progress, not future calendar dates.

The narrow, upward envelope follows a closed-trade sample with relatively little realized downside. P5, median and P95 full-horizon net profits are USD 218.08, USD 244.43 and USD 274.87. All 1,000 resampled paths ended profitable and no breach of the USD 700 floor was observed.

The 0.14% P95 closed-trade drawdown contrasts with the 14.46% historical equity drawdown. A basket that recovers from a substantial floating loss and closes profitably enters this bootstrap mainly as its recorded closing outcomes. The band consequently cannot recreate the March 2020 equity dip, establish a guaranteed future win rate or validate execution despite the five critical messages.

MC profit: P5 $218.08 · median $244.43 · P95 $274.87. Profitable 100.0% · USD 700 floor breach 0.0%.

MC statistics and plotted band: matched original cTrader interface JSON export for this run.

A pointwise band does not give a 90% chance that an entire future path stays inside it. Floor breaches are finite-model frequencies, not account-ruin probabilities. Closed-trade resampling omits intra-basket floating losses and broker stop-out.

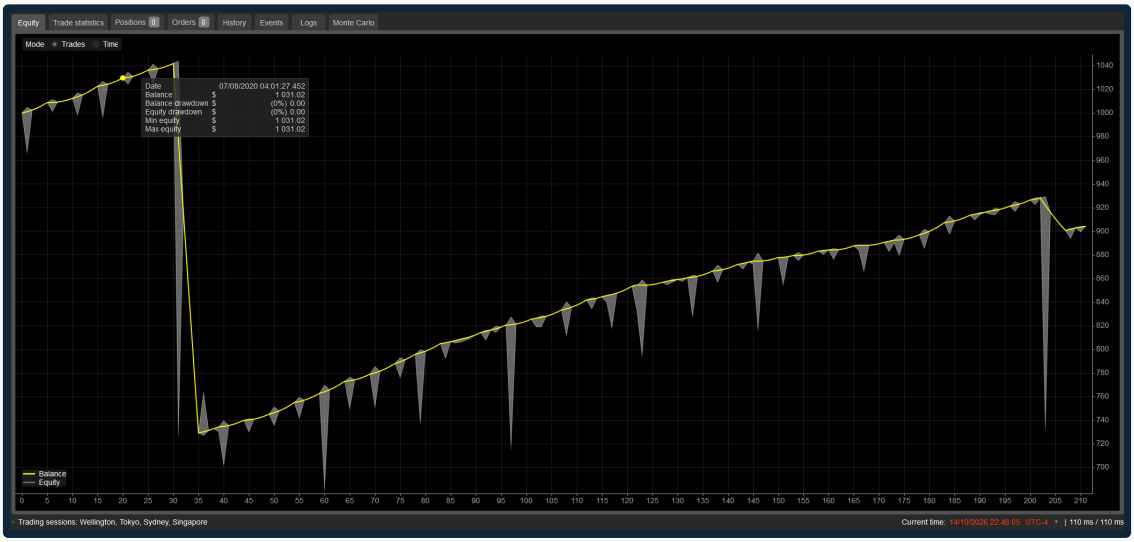
PAIR PROFILE

EURCAD

H1 · 1 January 2020–14 October 2026 · Historical simulation

NET PROFIT	CUMULATIVE RETURN	MAX. EQUITY DD
−$95.90	-9.59%	34.54%

HISTORICAL BALANCE AND EQUITY



cTrader capture: EURCAD.PNG. Yellow = balance; grey = floating equity. Axis = closed trades; the dated discussion uses native timestamps.

The large early balance step is a realized five-buy-leg loss of −USD 312.85 on 27 October 2021. The deepest cumulative equity trough came later, on 29 April 2022: USD 683.30 against the earlier sampled peak of USD 1,043.91 on 21 April 2021, a 34.54% drawdown.

That later trough includes the realized 2021 loss plus approximately USD 79.19 of floating loss in another five-leg buy basket. The later basket closed at +USD 10.22 on 6 May 2022, but the account never regained its old equity peak within the study. Subsequent accumulation was interrupted by a further −USD 27.63 basket close on 3 March 2026. Final equity was USD 904.10, preserving the study's −9.59% cumulative return.

Basket outcome and time at risk	Archived measurement
All closed baskets / mean net	62 / −$1.55
Median / P95 / longest holding time	0.23 / 21.42 / 236.80 calendar days
Time in market / baskets reaching five legs	22.22% / 48.39%

Initial $1,000.00 · final equity $904.10 · 211 closed legs · leg win rate 92.89% · leg PF 0.72.

Evidence note: 1 critical log message (EntityNotFound).

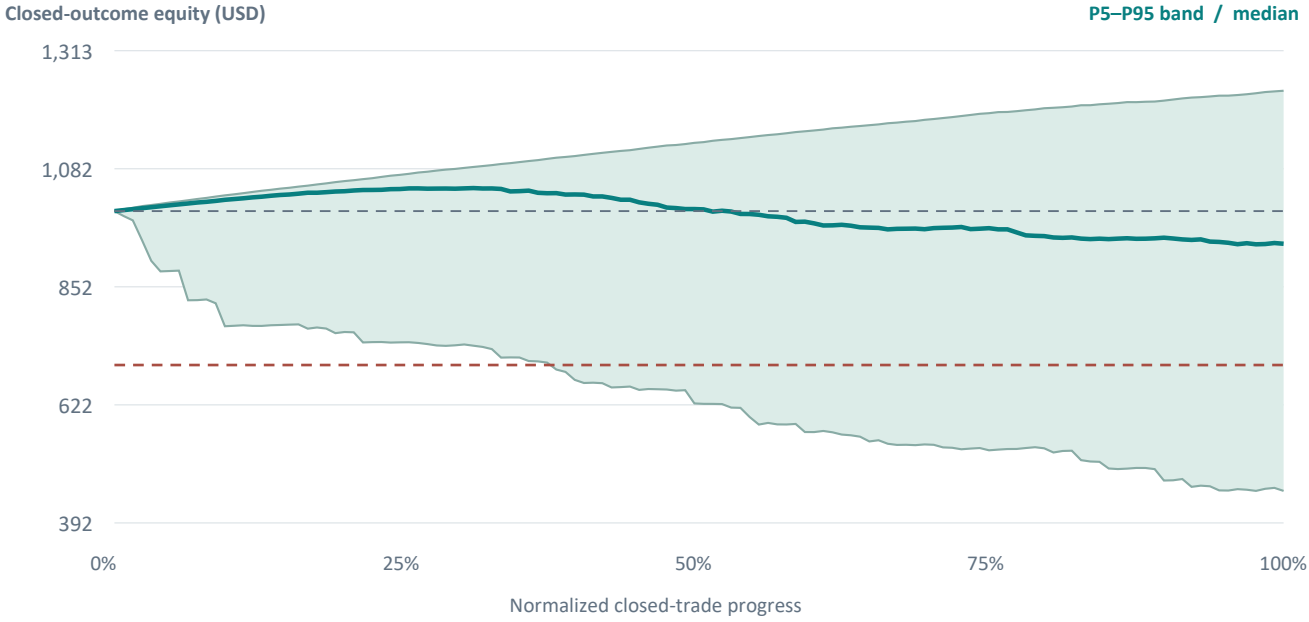
All-closed basket measurements above include shutdowns. One isolated historical test; run ID: b121d8b732b7. Calendar exposure is not a margin or leverage measurement.

PAIR ANALYSIS / MONTE CARLO

EURCAD · resampled outcomes

Matched closed-trade BlockBootstrap · 1,000 iterations · Five-trade blocks · Seed 0

P5 NET PROFIT	MEDIAN NET PROFIT	P95 NET PROFIT
−$545.34	−$63.68	$234.55



Teal line = median; shaded band = pointwise P5–P95 equity; grey dashed line = USD 1,000 start; red dashed line = USD 700 floor. The 128 plotted samples are interpolated over trade progress, not future calendar dates.

The lower envelope falls substantially while the median ends below starting capital. Full-horizon P5, median and P95 net profits are −USD 545.34, −USD 63.68 and USD 234.55. Only 39.7% of paths end profitable, and 27.2% breach the USD 700 floor. The positive upper tail does not outweigh the negative central outcome in this sample.

The matched local-engine calculation resamples the recorded large loss and smaller outcomes; it does not identify a new profitable configuration. P95 closed-trade drawdown is 61.21%. Floating basket trajectories and broker stop-out are absent, so neither the band nor the breach count should be interpreted as a complete measure of account-loss risk.

MC profit: P5 −$545.34 · median −$63.68 · P95 $234.55. Profitable 39.7% · USD 700 floor breach 27.2%.

MC statistics: local cTrader engine calculation from this run's closed trades; not an interface JSON export.

A pointwise band does not give a 90% chance that an entire future path stays inside it. Floor breaches are finite-model frequencies, not account-ruin probabilities. Closed-trade resampling omits intra-basket floating losses and broker stop-out.

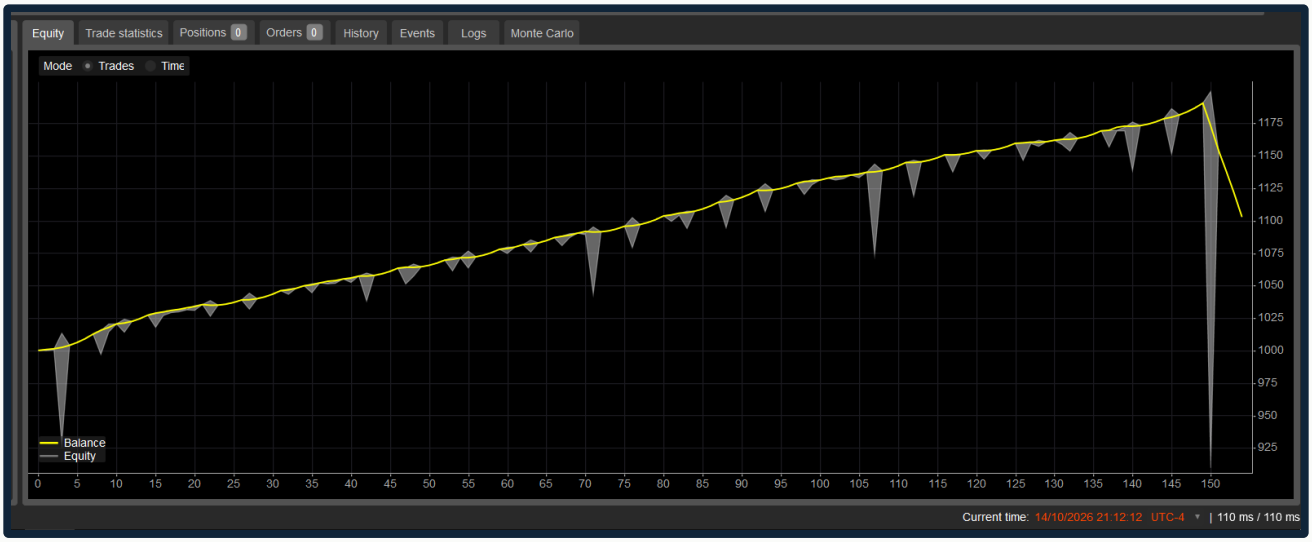
PAIR PROFILE

CADCHF

H1 · 1 January 2020–14 October 2026 · Historical simulation

NET PROFIT	CUMULATIVE RETURN	MAX. EQUITY DD
+$102.91	10.29%	23.83%

HISTORICAL BALANCE AND EQUITY



cTrader capture: CADCHF.PNG. Yellow = balance; grey = floating equity. Axis = closed trades; the dated discussion uses native timestamps.

The yellow balance accumulates gains while a prolonged final buy basket creates a deep grey equity excursion. Maximum drawdown was 23.83%, from a sampled USD 1,199.29 peak on 7 April 2025 to USD 913.53 on 17 October 2025. Five buy legs were exposed; the basket had first opened on 3 April 2025.

That basket remained open until shutdown on 14 October 2026, when it realized −USD 87.49. Its long lifetime is not apparent from equal closed-trade spacing on the screenshot. The terminal balance decline is a test-end liquidation, not an established stop-out. The old equity peak remained unrecovered and final equity was USD 1,102.91. Completed-basket statistics exclude the shutdown basket even though total profit includes its loss.

Basket outcome and time at risk	Archived measurement
All closed baskets / mean net	56 / $1.84
Median / P95 / longest holding time	0.20 / 9.91 / 559.33 calendar days
Time in market / baskets reaching five legs	25.57% / 32.14%

Initial $1,000.00 · final equity $1,102.91 · 154 closed legs · leg win rate 90.26% · leg PF 2.15.

Evidence note: one end-of-run basket excluded from completed-basket statistics.

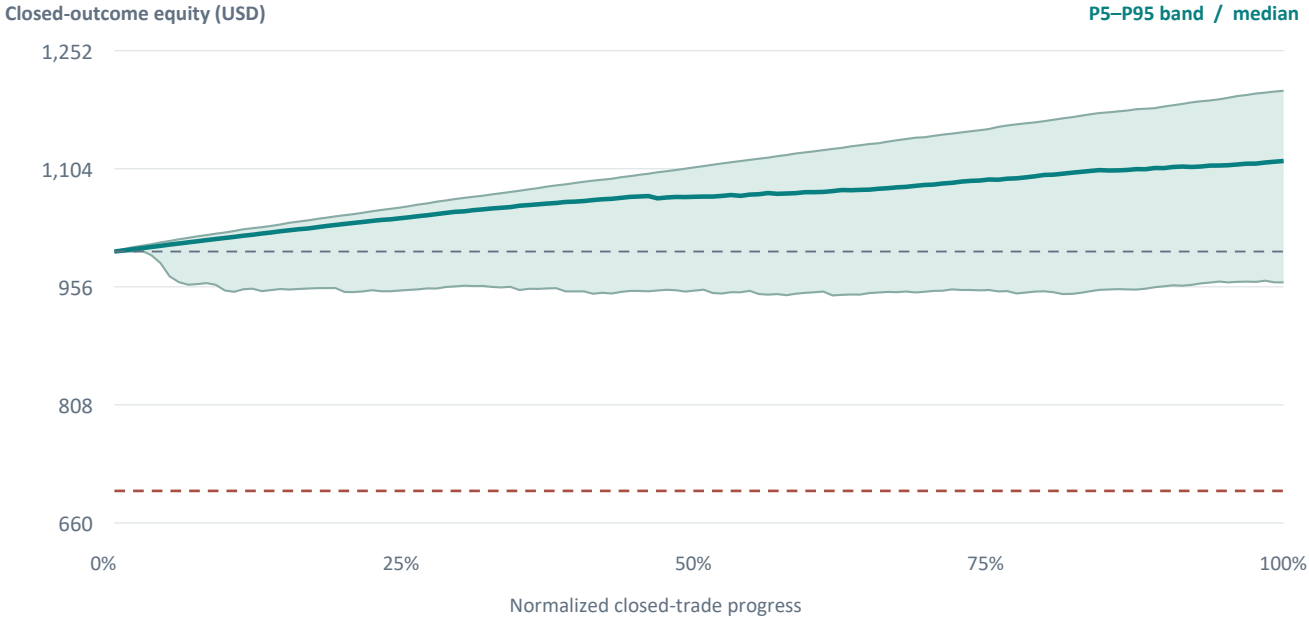
All-closed basket measurements above include shutdowns. One isolated historical test; run ID: ac1266808a5e. Calendar exposure is not a margin or leverage measurement.

PAIR ANALYSIS / MONTE CARLO

CADCHF · resampled outcomes

Matched closed-trade BlockBootstrap · 1,000 iterations · Five-trade blocks · Seed 0

P5 NET PROFIT	MEDIAN NET PROFIT	P95 NET PROFIT
−$38.64	$113.46	$201.52



Teal line = median; shaded band = pointwise P5–P95 equity; grey dashed line = USD 1,000 start; red dashed line = USD 700 floor. The 128 plotted samples are interpolated over trade progress, not future calendar dates.

The envelope widens around a rising median, with a lower endpoint below the original USD 1,000 capital. P5, median and P95 full-horizon net profits are −USD 38.64, USD 113.46 and USD 201.52. Profitable endings occur in 89.6% of paths; no USD 700 floor breach was observed in 1,000 iterations.

This matched local-engine calculation includes the terminal shutdown loss in the closed-trade sample. P95 closed-trade drawdown is 13.02%, compared with 23.83% historical equity drawdown. Resampling does not preserve the final basket's approximately eighteen-month calendar exposure or its floating path. Zero observed floor breaches therefore does not show that the prolonged basket could be held safely at higher volume.

MC profit: P5 −$38.64 · median $113.46 · P95 $201.52. Profitable 89.6% · USD 700 floor breach 0.0%.

MC statistics: local cTrader engine calculation from this run's closed trades; not an interface JSON export.

A pointwise band does not give a 90% chance that an entire future path stays inside it. Floor breaches are finite-model frequencies, not account-ruin probabilities. Closed-trade resampling omits intra-basket floating losses and broker stop-out.

CHART CONTEXT / 1 OF 2

What caused the sharp 2020 drops?

The records identify the exposed baskets and their exits. Primary market sources explain the conditions surrounding them.

The March 2020 episodes coincide with the COVID-19 shock. The Reserve Bank of Australia documents rapid currency moves, strong US-dollar demand and impaired FX liquidity. A buy grid exposed to a continuing decline could fill every permitted leg while losses kept growing. The five-pip step controls additions; it does not limit the subsequent adverse move.

Pair / date UTC	Observed event	What the trading records show
AUDUSD 18 March 2020	Realized basket loss **–USD 301.57**	Five buy legs closed at 15:24 UTC. This lowered realized balance; it was not merely a grey floating-equity spike.
NZDUSD 19 March 2020	Realized basket loss **–USD 302.32**	Five buy legs closed at 00:55 UTC. The loss remained part of the account's later cumulative drawdown.
AUDCHF 19 March 2020	Floating equity DD **28.23%**	Five buy legs were exposed at the sampled low. That basket later closed **+USD 18.44** on 25 March.
EURGBP 23 March 2020	Floating equity DD **14.46%**	Five sell legs were exposed at the sampled low. That basket later closed **+USD 19.77** on 24 March.

Which control appears to have closed the losses?

AUDUSD and NZDUSD losses were close to 30% of their pre-basket balances, approximately USD 1,005 and USD 1,007. In each basket, all five positions closed through bot requests at one timestamp, away from their recorded server-stop prices. The basket losses were below the USD 700 money allocation. This is **consistent with the 30%-of-balance stop**; the saved logs do not preserve an explicit trigger message.

> **A later trough can include the earlier crash.** AUDUSD's largest cumulative equity drawdown occurs on 2020-09-01, and NZDUSD's on 2022-01-28. Both combine the March 2020 realized loss with a later floating loss. Those later dates should not be labeled a fresh COVID shock.

Market background: RBA, May 2020, Box B; Federal Reserve, USD liquidity measures, 19 March 2020. These sources support the market context; they do not measure this broker's spread or prove a sole cause for an individual fill.

CHART CONTEXT / 2 OF 2

Date the trough before naming the event

All thirteen largest equity drawdowns were reconstructed from the native report envelopes and matched to the original basket records.

Pair	Peak date	Trough date	DD %	Book / legs	Later basket result
EURUSD	2025-04-10	2025-04-21	12.98%	Sell / 5	+$10.17 · 2025-05-12
GBPUSD	2021-02-08	2021-02-24	24.86%	Sell / 5	+$1.34 · 2021-03-24
AUDUSD	2020-03-12	2020-09-01	34.70%	Sell / 5	+$0.98 · 2020-09-22
USDCAD	2024-11-26	2025-02-03	24.46%	Sell / 5	−$44.49 · 2025-04-10
USDCHF	2023-12-19	2023-12-28	15.75%	Buy / 5	+$18.68 · 2024-01-17
USDJPY	2025-10-21	2026-07-06	30.44%	Sell / 5	+$3.56 · 2026-07-06
AUDCHF	2020-03-16	2020-03-19	28.23%	Buy / 5	+$18.44 · 2020-03-25
EURAUD	2022-03-16	2022-04-05	27.11%	Buy / 5	−$14.58 · 2022-05-10
EURCHF	2026-07-06	2026-09-30	17.70%	Sell / 5	−$132.49 · 2026-10-14 Shutdown / EndOfRun
NZDUSD	2020-03-12	2022-01-28	36.84%	Buy / 5	−$2.30 · 2022-03-06
EURGBP	2020-03-23	2020-03-23	14.46%	Sell / 5	+$19.77 · 2020-03-24
EURCAD	2021-04-21	2022-04-29	34.54%	Buy / 5	+$10.22 · 2022-05-06
CADCHF	2025-04-07	2025-10-17	23.83%	Buy / 5	−$87.49 · 2026-10-14 Shutdown / EndOfRun

Read the dates correctly: UTC labels mark the end of sampled intervals, usually one hour; the exact extremum tick is unavailable. The final column is the outcome of the basket open at the trough, not recovery of the account's full drawdown. Prior realized losses also contribute in AUDUSD, NZDUSD, EURCAD and USDJPY.

EURUSD, April 2025: a sell basket faced euro appreciation amid trade-policy repricing; see ECB Bulletin 4/2025, section 4. **USDCAD, February 2025:** its sell basket had been open since November 2024; Bank of Canada and CFEC minutes document CAD weakness, tariff uncertainty and volatility. These are contextual associations, not sole-cause findings.

> For other dates, the records establish adverse grid exposure but do not isolate one macroeconomic cause. EURCHF and CADCHF's final negative baskets were **shutdown liquidations**. Full intervals, recovery-touch dates and primary references are supplied in **analysis-supplement**.

CURRENCY BUCKETS / 1 OF 2

Which pairs had the highest ROI?

Historical reference: 1 January 2020–14 October 2026. Source build: SGU v2.2 AB A, assembly 2.2.2.0.

For a shortlist of **at least two distinct pairs** with equal funding, the highest historical ROI choice is **GBPUSD + USDCHF**. They are the two highest individual ROI results, so their mean exceeds every larger equal-funded combination. This is a retrospective selection from the same history, not a forecast of the best future pair mix.

CUMULATIVE ROI	ANNUALIZED*	NET PROFIT
54.24%	6.60%	+$1,084.72

Funding: USD 1,000 per separate account, USD 2,000 total. The unchanged historical paths finish at **USD 3,084.72**. Both retain fixed 0.01 lots per leg, five legs and the documented risk settings. The capital assumption is essential to the return calculation.

Bucket / pairs	Start USD	Net USD	Final USD	ROI	Ann.*
GBPUSD Single-pair benchmark	1,000	565.79	1,565.79	56.58%	6.83%
GBPUSD + USDCHF Selected for historical ROI	2,000	1,084.72	3,084.72	54.24%	6.60%
GBPUSD + USDCHF + EURUSD	3,000	1,550.21	4,550.21	51.67%	6.33%
GBPUSD + USDCHF + EURUSD + EURAUD	4,000	1,866.84	5,866.84	46.67%	5.81%
GBPUSD + EURUSD No critical messages in these runs	2,000	1,031.28	3,031.28	51.56%	6.32%
All thirteen documented pairs	13,000	3,083.03	16,083.03	23.72%	3.19%

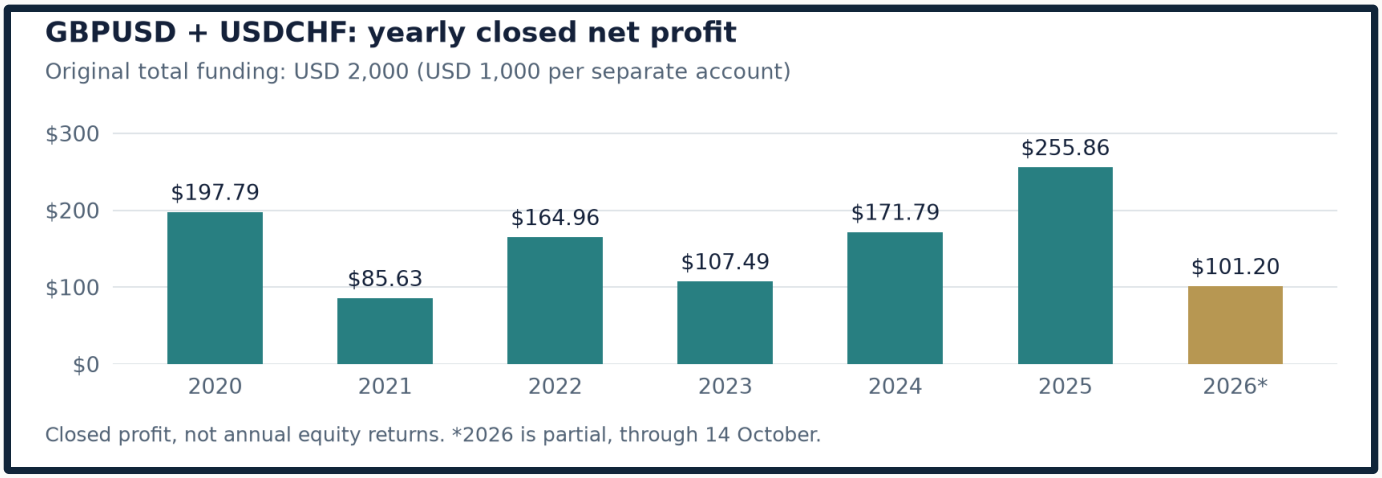
*Annualized endpoint growth = (final / initial)$^{365.2425 / 2478}$ − 1, over approximately 6.78 years. It summarizes endpoints; position sizes were not compounded and yearly gains were not smooth. Profits include recorded commission and swap, but not tax, hosting or every future execution cost.

Execution qualification: USDCHF's source run contains **39 critical EntityNotFound messages**. Reconcile them and rerun the intended operating package before deployment. GBPUSD + EURUSD is the higher-ROI two-pair comparison among runs without recorded critical messages; this does not establish future safety.

CURRENCY BUCKETS / 2 OF 2

Use the bucket with explicit funding

The historical return is a reference outcome. A reliable expected live return is not estimated from this selected sample.



Yearly totals allocate exported net profit to each trade's closure year. They exclude floating P/L and are not annual mark-to-market returns.

Reading the bars: 2025 contributes the most closed profit (USD 255.86); 2021 is the lowest full year (USD 85.63). The gold 2026 bar is only USD 101.20 through 14 October. A flat or positive yearly closed total can coexist with a large open loss; a long-held basket's entire closing result falls into its exit year. The varying bars do not establish a steady annual return.

Checkpoint	Application to GBPUSD + USDCHF
Funding model	Two separately funded USD 1,000 accounts; no transfers, rebalancing or volume scaling. USD 1,000 shared across both does not reproduce this illustration.
Floating risk	Standalone maximum equity DD: GBPUSD 24.86%; USDCHF 15.75%. Exact combined equity drawdown is unmeasured: native points contain interval extrema, not synchronized marks.
Currency exposure	Both pairs involve USD. Actual long/short positions can reinforce or offset exposure; two symbols do not establish diversification. Ten 0.01-lot legs total 0.10 gross lots, not net FX risk.
Shared account	The 30% basket stop reads current Account.Balance. Other symbols' profits/losses and common margin can change exits. Obtain a true simultaneous replay or forward record for that account.

Before operating the shortlist

1. Load the two matching study files on stopped local demo instances at H1; preserve signal and risk settings. **2.** Review ownership and choose distinct labels for new books. **3.** Resolve USDCHF errors and validate the intended normal package. **4.** Freeze the shortlist, collect new forward observations, and record combined and per-pair equity at common timestamps.

The standalone maximum dollar drawdowns sum to **USD 474.10**: a historical upper bound for the sum of unchanged separate-account paths, **not observed joint drawdown or a future loss limit**. CSV comparisons, yearly totals, dated episodes and the standalone chart are in **analysis-supplement**.

POSITION SIZING / BASELINE AND SENSITIVITY

Interpret the baseline before scaling

The study uses fixed volume and common indicator settings. Larger positions change both opportunity and risk.

Each separate USD 1,000 account used **0.01 lots per leg**, up to five legs, without increasing volume as the balance grew. Native reports record **1:20 account leverage**. That margin setting alone does not multiply trading P/L when executed volume, prices and costs stay unchanged.

Position size is not the leverage ratio. The saved FX contract is 100,000 base-currency units per lot. At USD 1,000 equity, one 0.01-lot USDCHF leg represents USD 1,000 notional; five legs represent USD 5,000, about 5× equity. For GBPUSD at 1.30, five legs are about USD 6,500. Exposure/equity changes with prices and floating P/L. The results should therefore not be labeled unleveraged or 1× market exposure.

A sizing sensitivity, not another backtest

For the **GBPUSD + USDCHF** bucket, keep original funding at **USD 2,000**. Define **s = proposed lots per leg / 0.01**. If all trades, fills, exits and proportional costs were unchanged, net profit would be approximately s × USD 1,084.72; cumulative ROI would be s × 54.236%. The rows below show only that arithmetic over the full 2020–2026 period.

Size multiple	Lots / leg	Net USD*	ROI*	Evidence
1× original size	0.01	1,084.72	54.24%	Archived baseline
2× original size	0.02	2,169.44	108.47%	Arithmetic only
3× original size	0.03	3,254.16	162.71%	Arithmetic only

*At 2× size, an unchanged USD 100 loss also becomes USD 200. Annualized endpoint sensitivity = $(1 + s \times 0.54236)^{365.2425 / 2478} - 1$; do not multiply the 6.60% annualized baseline by s. The companion spreadsheet supplies these formulas and an editable size input.

> **Actual SGU paths will not generally scale linearly.** The 30%-of-current-balance stop, fixed USD 700 allocation, broker-stop distance and margin can change entries and exits. Larger positions increase loss sensitivity; trends, gaps, repeated stopped baskets or failed exits can cause **loss of the entire account**. No setting guarantees a total-loss ceiling. Rerun the proposed size with unchanged risk rules and validate it forward on demo; do not infer a ruin probability from this table.

Optimization status: the publisher reports no indicator fine-tuning for this study; identical saved settings confirm consistency, not the absence of all earlier research. The bucket was selected retrospectively. Better returns from tuning are unproven. Limit degrees of freedom as explained on pages 50–53.

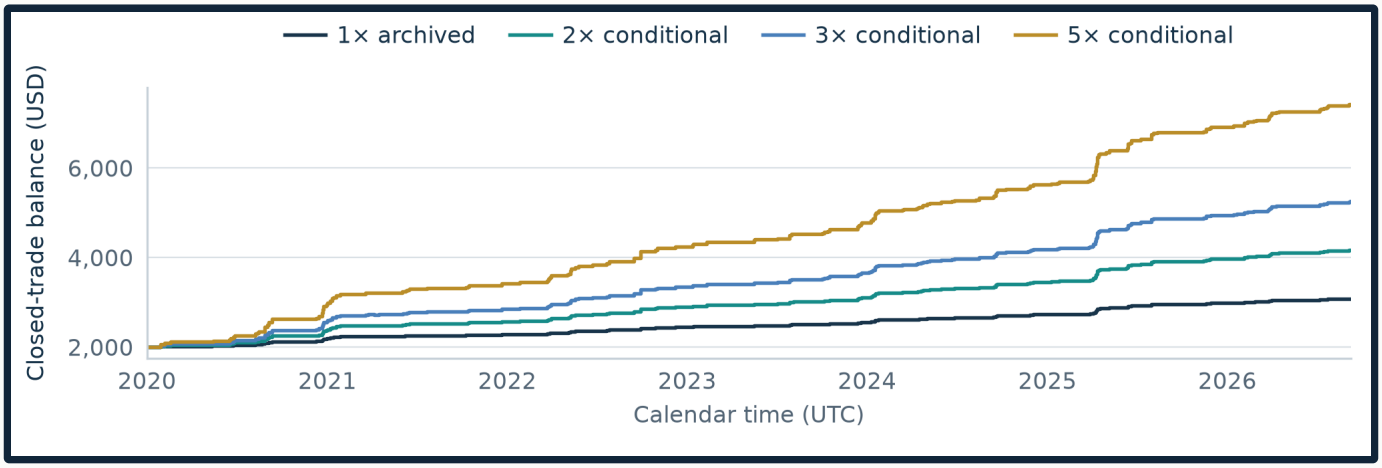
Companion: **analysis-supplement / position-size-sensitivity-calculator.xlsx**. Official references: cTrader leverage and margin · Trading conditions / stop-out.

LEVERAGE SCENARIOS / 1 OF 2

What larger positions would change

GBPUSD + USDCHF · Two separate USD 1,000 accounts · Frozen-path simulation, 2020–2026

To illustrate greater market exposure, multiply each original 0.01-lot leg by **s**, while keeping total starting cash at USD 2,000. The model scales archived trade P/L, commission, swap and floating-equity changes. It assumes every original entry and exit remains unchanged; it does not rerun SGU's trading decisions, stops or broker margin.



Chronological sum of closed-trade balances, starting at USD 2,000 across two separate accounts. Floating losses are absent.

The lines share the same rises and pauses because they reuse the same closed trades. Larger size stretches each gain or loss vertically; it creates no new trading edge. Their smooth appearance hides the open-position losses quantified on the next page. A positive ending balance alone does not establish a feasible path.

Position size	Lots / leg	Net USD*	Final USD*	ROI*	Ann.*
1× original	0.01	1,084.72	3,084.72	54.24%	6.60%
2× original	0.02	2,169.44	4,169.44	108.47%	11.44%
3× original	0.03	3,254.16	5,254.16	162.71%	15.30%
5× original	0.05	5,423.60	7,423.60	271.18%	21.33%

*Only 1× is the archived backtest. Other rows are conditional frozen-path outputs, not executable backtests or forecasts. Annualized endpoint arithmetic = $(1 + s \times 0.54236)^{365.2425 / 2478} - 1$; the 6.60% baseline annual figure is not multiplied by s. Fixed lot size does not demonstrate reinvestment.

The original risk rules invalidate simple extrapolation. All enlarged GBPUSD scenarios reach sampled floating losses inconsistent with retaining the original 30%-of-current-balance stop. A real run would generally exit differently, potentially much earlier. USD 700 allocation, automatic broker stops, margin and execution add further differences. The higher return figures must be read with the loss and feasibility analysis on page 47.

Exportable chart and calculation rows: **analysis-supplement / leverage-path-scenarios.pdf**, **.svg** and **.csv**. Definitions and assumptions: **leverage-scenario-notes.md**. Size multiples are not broker leverage ratios.

LEVERAGE SCENARIOS / 2 OF 2

Test the losses and financing first

The same frozen paths show larger drawdowns. Separate-account minima are not a synchronized bucket drawdown.

Size	GBPUSD min. equity	GBPUSD max. DD	USDCHF min. equity	USDCHF max. DD
1×	$850.19	24.86%	$935.12	15.75%
2×	$700.38	44.54%	$870.24	26.61%
3×	$550.57	60.51%	$805.36	34.56%
5×	$250.95	84.86%	$675.60	45.41%

Each component starts at USD 1,000. At 5× size, GBPUSD's sampled equity falls to **USD 250.95**, with **84.86%** peak-to-trough drawdown. Its minimum occurs in February 2021; USDCHF's minimum occurs in January 2020. Drawdown is measured from each changing prior equity peak, so percentage drawdown is not simply the original percentage multiplied by size.

Observed conflicts with the balance stop: the 2× GBPUSD path already exceeds the original 30%-balance loss trigger in the sampled interval ending **22 February 2021, 16:00 UTC**. The 5× USDCHF path does so on **16 January 2020, 13:00 UTC**. These are incompatibility checks on stable-balance intervals, not exact simulated stop-fill times; other constraints can intervene earlier.

Broker leverage changes the margin requirement

Illustration: five 0.05-lot USDCHF legs represent USD 25,000 base notional. With uniform leverage, their indicative margin is notional / leverage. The P/L for the same executed exposure and price path stays the same.

Assumed broker leverage	Indicative margin	Interpretation at USD 1,000 equity
1:20	USD 1,250	Exceeds starting equity before any floating loss.
1:50	USD 500	Leaves less capacity for adverse open P/L.
1:100	USD 250	Less collateral required; the same market loss remains.

Full-account-loss stress: at 7× original size, GBPUSD's frozen path crosses zero in the interval ending 24 February 2021, 02:00 UTC, with a mathematical minimum of −USD 48.67. A later paper recovery cannot finance trading through zero. No ending-return figure is published for that path. This is not an estimate of ruin probability, an available leverage recommendation or a prediction of broker liability.

Actual symbol tiers, conversion, free margin and broker stop-out differ; the three margin ratios are illustrative assumptions. Confirm broker conditions, rerun the proposed volume with all intended protections, then assess forward demo equity. Larger positions can cause loss of the entire account. Read the cTrader margin reference and stop-out conditions.

GETTING STARTED

Install and identify the product

Begin with a demo account and the exact package and settings you intend to operate.

STEP 1

Install and verify the operating package

Open SGU Smart Grid Ultimate.algo in cTrader. Confirm assembly 2.3.0.0 / revision 2.3.0-no-adx.2 and build no-adx-20261018.2. Keep the original package and start a fresh test to load the update.

STEP 2

Choose the right account

Sign in to the intended cTrader account and verify the broker, deposit currency, account number and hedging mode. The study's monetary parameters are USD amounts because its deposit currency is USD.

STEP 3

Add a local instance

Select the cBot, add an instance and choose the intended symbol and timeframe. For a study-style setup, select H1. Use local Windows or Mac execution; SGU Cloud operation is exit-only.

STEP 4

Load and review the saved settings

Keep the instance stopped. Use Modify parameters / Open. Current 56-value presets are in settings-2.3; original 61-value records remain in study-settings. Check symbol, period and all values; preserve the tag when recovering an existing book.

STEP 5

Validate, then start on demo

Check leg size, leg count, loss settings and ownership. Save the reviewed parameters, reproduce the backtest, then start the local demo instance. Inspect logs, actual volume and accepted stops when exposure first opens.

> **Volume check:** a requested size below the broker minimum is raised to that minimum. Confirm the actual order size and full five-leg exposure. Platform reference: Start a cBot.

CONFIGURATION AND VALIDATION

Load settings, then reproduce the test

A .cbotset file stores an instance's parameters. Backtest environment settings must also be checked.

Import the matching study settings

For 2.3, use **Modify parameters → Open** and the pair file in **settings-2.3**, such as **SGU-v2.3-EURUSD-H1-Study700-migrated.cbotset**. It removes the five ADX keys; its historical source remains in study-settings. Keep the instance stopped and verify symbol, H1 chart and all retained values.

All thirteen archived files retain **Instance Tag = SGU_EURUSD**; the selected symbol determines the market. Preserve that tag for exact reproduction or recovery. For a new independent book, review ownership before choosing a distinct tag. Older presets can acquire defaults for newer parameters.

Recreate the documented environment

Check	Study reference
Dates and data	1 January 2020–14 October 2026 UTC; server tick data. Preserve the actual data availability and stop time reported by the platform.
Capital and leverage	USD 1,000 for all thirteen actual backtests; leverage 1:20.
Costs	USD 30 per million USD volume. The native spread field is unreported; a 2-pip spread guard is not evidence of a fixed test spread.
Trading parameters	Current migrated file: 56 values, USD 700 allocation. The original archived file has 61 values. Migration does not create a new 2.3 result. "Money500" is a different configuration.
Output records	Save the native report, parameters, run identity, actual settings, equity drawdown and execution log. Retain any unfinished/end-of-run basket context.

Run and inspect the backtest

1. Select the instance's **Backtesting** tab. **2.** Open the settings cog and enter the capital, commission, tick-data and spread assumptions; set the date range. **3.** Click **Play**; use Visual mode for a short behavior check if needed. **4.** Inspect equity, trades and logs, then save the report and parameters.

> **Build identity:** exact historical reproduction uses **SGU v2.2 AB A, assembly 2.2.2.0** and its 61 values. Current operation uses **SGU 2.3, assembly 2.3.0.0** and 56 values. Validate the migrated configuration separately; it has no new performance record here.

Official references: Using .cbotset files · Backtest a cBot.

OPTIMIZATION / 1 OF 4

Give the search fewer choices

Avoid too many degrees of freedom. Overfitting (curve-fitting) means selecting rules that fit historical noise and fail to generalize.

Degrees of freedom are the choices you allow the research process to make: parameter values, timeframes, symbols, date windows and selection rules. The 56 available settings are not 56 independent research variables. A fixed tag or disabled alert does not add a trading hypothesis.

A winner can emerge by chance when many alternatives are tried on the same history. Manual tweaks, discarded runs and later search rounds still count. A genetic algorithm can reduce computation; it does not remove selection bias. There is no universal safe parameter count or trade-to-parameter ratio.

Freeze the operating and risk constraints first

Parameter family	Treatment for a controlled SGU study
Ownership and alerts	Keep Instance Tag, credentials and notification settings out of performance searches. Keep diagnostic collection consistent.
Capital and risk limits	Set lot size, leg cap, money allocation and protective stops from the intended risk constraints. Keep them fixed while comparing entry or target hypotheses.
Confirmation shape	Begin with fixed K / Slowing / D, smoothing and timeframe. If a change is justified, record it as a separate hypothesis.
Entry gates and targets	Vary a small, stated set. Confirmation On/Off, a short confirmation window or a modest ATR range can answer different questions; avoid releasing them all together.
Thresholds and sessions	Use a reasoned, bounded range if tested. Choosing levels, direction, trading hours or symbols after seeing results also adds selection freedom.

> Do not widen a stop simply because the historical profit improves. A calm sample may contain too few severe moves to assess the resulting tail risk. A changed risk limit creates a different strategy validation case.

Basis: earlier SGU manual, Calibration Protocol, PDF page 20; López de Prado, *Advances in Financial Machine Learning*, Chapter 11, printed pp. 151–157. The bounded workflow here is an editorial recommendation, not a proven optimum. Reading references: page 62.

OPTIMIZATION / 2 OF 4

Run one small experiment at a time

SGU 2.3 example: compare confirmation Off and On while keeping trading and risk settings fixed.

Two passes	Enable Confirmation	Purpose
A	Off	Primary stochastic only
B	On	Primary plus higher-timeframe stochastic

This is a **two-pass example**, not an optimized preset. Keep the H1 trigger, D3 confirmation shape, H1 ATR target, 5-pip spacing, 0.01 lots, five legs and USD 700 allocation unchanged. Fix all remaining values and the environment before running. ADX is absent from 2.3.

Check which settings are active

Switch / dependency	Consequence for the search
Confirmation Off	Confirmation timeframe, oscillator settings, levels and Valid For do not gate entries. Varying them adds duplicate behavior.
ATR target Off	ATR settings do not determine the normal target. With ATR On, the fixed Basket Target remains a fallback.
Use Chart Timeframe for ATR On	The chart supplies the ATR timeframe. Changing only the stored ATR Timeframe does not change that selection.
Guard Added Legs Off	Spread and weekend gates apply to first entries. Indicator permission remains a first-leg condition even when the added-leg guard is On.

With **2.3** loaded in a fresh test, open **Optimisation → Parameters** and select only **Enable Confirmation**, including both On/Off values. Exclude chart Timeframe and other fields. Choose **Algorithm → Grid**. Fix Criteria, dates, data, capital, costs and spread before Play. Save every pass and build identity; an .optset stores ranges, not the whole environment.

> Earlier four-pass and 72-pass examples included an ADX axis and used USD 500. They remain historical research files, not current 2.3 templates. Without ADX, three K values × three ATR multipliers × two spacings × two confirmation states make 36 nominal combinations, before excluding inactive dependencies.

Platform controls: cTrader · Optimize a cBot. Record subsequent sensitivity passes too, including repeated baseline values. A later search does not erase earlier trials. Each further selection must be logged.

OPTIMIZATION / 3 OF 4

Keep future evidence separate

Write the dates and decision rules before inspecting results. A reused holdout becomes development data.

DEVELOP	VALIDATE	FORWARD
Earlier history	**Later history**	**New observations**
Search within a declared budget.	Compare frozen shortlisted settings.	Evaluate the frozen decision.

1. Define the experiment

Record the hypothesis, candidate ranges, trial limit, dates, regime coverage, cost model and rejection rules. Fix the maximum acceptable equity loss and exposure before seeing the winning pass.

2. Develop on earlier observations

Retain every pass, including failures. Check ranges, trends, volatility changes and stress episodes. Years of history and raw trade counts alone do not establish a usable sample.

3. Test the shortlist on later observations

Freeze settings first. Use consistent assumptions; label deliberate cost stress separately. If later results cause another tweak or selection round, that period has influenced development and is no longer an untouched final test.

4. Observe the frozen setup forward

Use genuinely new observations or demo data after freezing the decision. Define a review schedule and intervention rules in advance. Keep the chosen build and settings with the forward record.

> **Walk-forward option:** repeat a fixed train-then-test rule through time and join only the later test segments. Account for basket overlap and indicator warm-up at boundaries. A flat restart changes the strategy path; carrying positions needs a documented ownership and scoring policy.

In financial cross-validation, purging removes overlapping training labels; embargo helps limit remaining temporal dependence. For SGU, assess baskets spanning a split across their full lifetime, rather than randomly separating legs. This guide does not claim a formal purged-CV or PBO study. AFML Ch. 7, printed pp. 104–109; Ch. 12, pp. 161–162.

OPTIMIZATION / 4 OF 4

Select for stability, then stress the result

Treat the optimizer's score as a screening tool. Review floating risk, basket outcomes and execution before accepting a candidate.

SGU's **Custom / GetFitness** rejects passes below Fitness Min Trades. The default threshold of 30 counts closed legs, not independent baskets. The score includes owned open-position P/L and uses equity drawdown with a small denominator floor. It is not a deployment approval rule.

Review	What a useful candidate should withstand
Full risk and outcome	Read realized net profit and open-position P/L, maximum equity drawdown, worst basket, basket count, duration, exposure and margin use. Preserve EndOfRun outcomes separately.
Nearby settings	Inspect a small, predeclared neighborhood. Favor a broad region of acceptable results over one isolated peak. Count any neighborhood used for selection in the trial log.
Market regimes	Break results into chronological subperiods, including sustained trends and volatility shocks. Explain whether profit depends on one period, pair or exceptional basket.
Costs and execution	Stress spread, commission, swap and slippage assumptions explicitly. Review rejected entries, stop changes, closing failures and unresolved exposure.
Resampling	Use basket-aware blocks if available. The published five-trade BlockBootstrap can split baskets and omits intrabasket floating paths. Resampling a selected trade list does not remove selection bias or create new out-of-sample evidence.
Forward record	Freeze the candidate, test on demo, and record actual fills, stops and recovery behavior. A profitable historical pass is not evidence of reliable live operation.

> **Why trade count matters:** one archived EURUSD validation contains 71 closed legs but only 25 completed baskets. Those are different units, and baskets can remain dependent across time. Neither 30 trades nor 30 baskets is a universal statistical sufficiency rule.

That local example uses a USD 500 budget, different settings and a changed spread model; its coverage review is marked pending. It illustrates recordkeeping, not a validated setting for this guide. The published 2020–2026 panel has already been inspected and cannot be relabeled untouched out-of-sample evidence.

Keep with each decision: all passes, hypothesis, ranges, build, broker/data, costs, dates, settings fingerprint, report, basket statistics, equity curve, logs and acceptance/rejection reason. Use the companion **optimization-templates** plan and trial log. AFML Ch. 11, printed pp. 151–157; sources: page 62.

DAILY OPERATION

Monitor the basket, not just the P/L

Keep the operating routine short, repeatable and tied to broker-confirmed state.

When	What to verify
Before starting	Correct account, symbol, timeframe, package version and tag; no competing owner; reviewed and saved settings; normalized volume; a functioning connection. Check accepted broker stops as positions open.
While running	Owned positions and pending orders, floating net loss, margin availability, server-stop status, basket state and the execution log. Review correlated exposure across other instances.
After a rejected or uncertain order	Check actual fills and remaining exposure at the broker. The recovery mechanism may retain an unresolved intention and block replacement entries. Do not force a duplicate order.
Before an upgrade	Prefer a flat account state for the affected book. Preserve original parameters, logs and recovery identity. Stop and confirm the resulting broker position/order state.
After a restart	Confirm the original ownership identity and that existing exposure is recognized. Review any recovery or ownership warning before permitting additional entries.
After a protective exit	Check that all owned legs and orders are closed or cancelled. A triggered stop or a pressed Stop button does not itself confirm a complete liquidation.

Optional notifications

The historical study has email, sound and Telegram disabled. Available alerts are supplementary and can fail or be delayed. They do not replace checking positions, logs and connectivity.

Telegram setup: configure while flat on desktop/demo. Create a bot with BotFather, open its chat and press Start, then obtain the numeric chat ID privately. Enter the token and ID in Telegram Alerts, enable Telegram and select Info for a startup check. Start the instance and verify receipt; Warning limits routine alerts. Plan any restart as a liquidation action. Telegram provides outbound alerts, not trading commands.

> Keep notification tokens and chat identifiers private. Native cTrader parameter files and reports can retain credentials even when custom research exports omit them. Remove secrets before sharing support evidence.

TROUBLESHOOTING

Resolve the cause before restarting

Use the instance log and broker state to distinguish strategy conditions from an execution problem.

Symptom	Likely checks and next action
No new trades	Check stochastic conditions, D3 history, allowed side, spread, UTC windows, ownership and recovery. Since 2.2.3, a confirmed margin refusal delays new entries up to 60 simulated minutes unless affordability improves; protection continues.
Netted-account refusal	Switch to an appropriate hedging account. Separate grid legs are required for the product's position-management model.
Ownership or recovery warning	Check every device for the same account, symbol and tag. Missing or damaged recovery records can block entries even while flat. Reconcile broker state before changing identity or local records.
Stop or entry rejected	Inspect the broker error, executable quote, minimum stop distance, volume rules and margin. Margin errors are aggregated since 2.2.3; Log Level = None suppresses routine logs, not critical safety diagnostics. An automatic stop still needs broker acceptance.
EntityNotFound in a log	Preserve the message with surrounding timestamps and position/order identifiers. Verify whether exposure or protection remains and request a review; do not classify a finished report as error-free.
Results differ from the guide	Compare build, currency/capital, data, dates, costs, spread, parameters and trades. The guide's study remains 2.2.2; retry timing since 2.2.3 can change trades. Match screenshots and Monte Carlo to the same run.
No notification received	Check for a disabled-channel warning, valid credentials/destination and desktop real-time operation. Review notification diagnostics privately. Plan a restart as liquidation of any owned exposure.

Send a useful support request

Include the product/build version, cTrader version, broker, hedging mode, symbol, timeframe, Instance Tag, time of the event with timezone, redacted parameters, log excerpt and the broker-confirmed position/order state. For backtests, attach the native report and run ID.

> Support: cg.walter.ma@gmail.com · +56 9 3366 9343. Never send passwords, API tokens or unredacted notification credentials.

SETTINGS REFERENCE · 1 OF 4

Ownership, exposure and protection

The 56 settings retained in 2.3, shown with historical study values; these are not universal defaults.

Setting / internal name	Study value	Meaning
Instance Tag InstanceTag	SGU_EURUSD	Position ownership label. Preserve across recovery; use distinct tags for separate instances on the same symbol.
Lot Size VolumeInLots	0.01	Requested size of each leg. Check broker-normalized size; maximum intended exposure is leg size times Max Legs.
Direction Side	Both	Allows either Buy or Sell. Only one directional basket is active at a time.
Vertical Spacing (pips) StepPips	5	Adverse price distance from the latest filled leg before another leg may open.
Max Legs MaxActivePositions	5	Maximum positions in a basket. Five 0.01-lot legs imply up to 0.05 intended lots.
Max Basket Loss (money) MaxBasketLossMoney	700	Mandatory basket loss budget in account currency; reaching it requests liquidation, without guaranteeing the final loss.
Basket Stop (pips, 0 = off) StopLossPips	0	Additional basket loss-distance trigger. Zero disables this optional pip trigger.
Basket Stop (% balance, 0 = off) StopLossPercent	30%	Additional basket net-loss trigger as a share of current account balance; not a maximum-equity-drawdown guarantee.
Broker Leg Stop (pips, 0 = auto) BrokerStopLossPips	0 (automatic)	Zero uses the budget-derived server stop. A positive value caps that distance if tighter; broker acceptance still applies.
Max Spread (pips, 0 = off) MaxSpreadPips	2	Blocks starting a basket above this spread. Later legs use it only when Guard Added Legs is on.
Max Slippage (pips, 0 = off) MaxSlippagePips	0	Zero uses market orders. A positive value requests an entry market-range tolerance.
Guard Added Legs GuardAddedLegs	Off	Off leaves later legs outside the spread and weekend entry guards; indicator gates remain first-leg only.
Log Level Verbosity	None	Log threshold. None suppresses routine logs in backtests/optimization; critical safety diagnostics remain unconditional.
Fitness Min Trades FitnessMinTrades	30	Minimum closed legs for an optimization score. Does not alter a single backtest or live trading.

The historical study has 61 archived values; SGU 2.3 retains 56 and removes the five ADX keys. Retained study values are identical across the thirteen native files, including Instance Tag = SGU_EURUSD. The selected symbol controls the market traded. Preserve an existing tag for recovery; assign distinct tags for separate books on the same symbol.

SETTINGS REFERENCE · 2 OF 4

Basket targets

The 56 settings retained in 2.3, shown with historical study values; these are not universal defaults.

Setting / internal name	Study value	Meaning
Basket Target (pips) TargetProfitPips	30	Fixed target from the weighted average entry; fallback when ATR target data is unavailable.
Enable ATR Target UseAtrTarget	On	Uses valid ATR from the last completed ATR bar; refreshes on each new ATR bar and can change during a basket.
Use Chart Timeframe for ATR UseChartTimeFrameForAtr	Off	Off selects the explicit ATR Timeframe below rather than automatically following the chart.
ATR Timeframe AtrTimeFrame	H1	Explicit series used for ATR while Use Chart Timeframe for ATR is off.
ATR Periods AtrPeriods	14	Lookback for the target volatility calculation.
ATR Smoothing AtrSmoothing	Exponential	Exponential averaging for the ATR series.
ATR Multiplier AtrMultiplier	1	Multiplies ATR in pips before the target bounds are applied.
Min Target (pips, 0 = off) AtrMinPips	2	Lower bound of the ATR target; zero would disable this bound.
Max Target (pips, 0 = off) AtrMaxPips	30	Upper bound of the ATR target; zero would disable this bound.

> The study enables the ATR target. Targets are measured from the weighted average entry; final net profit depends on commission, swap and execution. Historical ADX settings are archived, not exposed in 2.3: Off, D3, 14 periods, maximum 25, Block.

SETTINGS REFERENCE · 3 OF 4

Entry timing and confirmation

The 56 settings retained in 2.3, shown with historical study values; these are not universal defaults.

Setting / internal name	Study value	Meaning
Enable Stochastic Trigger UseStochasticTrigger	On	On requires a qualifying cross and extreme-zone reading. Off removes this gate; entries can still occur under the remaining rules.
Timeframe StochTimeFrame	H1	Series used for the entry trigger, which reads completed bars.
K Periods StochKPeriods	21	Trigger stochastic lookback.
K Slowing StochKSlowing	3	Smoothing applied to trigger %K.
D Periods StochDPeriods	3	Trigger %D signal-line averaging period.
Smoothing Smoothing	Wilder	Wilder averaging used by the trigger stochastic.
Oversold Level LowerThreshold	30	Buy requires the completed-bar upward cross with %K at or below this level.
Overbought Level UpperThreshold	70	Sell requires the completed-bar downward cross with %K at or above this level.
Enable Confirmation UseConfirmStochastic	On	Adds a slower stochastic permission gate for the first leg.
Timeframe ConfirmTimeFrame	D3 (three-day)	Three-day series used for the confirmation gate.
K Periods ConfirmKPeriods	21	Confirmation stochastic lookback.
K Slowing ConfirmKSlowing	3	Smoothing applied to confirmation %K.
D Periods ConfirmDPeriods	3	Confirmation %D signal-line averaging period.
Smoothing ConfirmSmoothing	Wilder	Wilder averaging used by the confirmation stochastic.
Oversold Level ConfirmLowerThreshold	30	Buy confirmation zone. Completed %K must remain inside it and a qualifying cross must exist.
Overbought Level ConfirmUpperThreshold	70	Sell confirmation zone, mirrored from the Buy confirmation condition.
Valid For (bars) ConfirmWindowBars	4	Candidate cross window includes forming-bar index 0; four considers indices 0 through 3 where available.

> The H1 trigger uses the latest completed bar. The D3 confirmation requires the completed %K to remain in the relevant extreme zone and searches a four-candidate-bar window that includes the forming bar. A forming-bar crossover can change before closure.

SETTINGS REFERENCE · 4 OF 4

Sessions, alerts and notifications

The 56 settings retained in 2.3, shown with historical study values; these are not universal defaults.

Setting / internal name	Study value	Meaning
Block Friday BlockFriday	On	Blocks starting baskets during the Friday UTC window; does not close existing exposure.
Friday From (UTC hour) FridayFrom	20	Inclusive Friday blocked-window starting hour in UTC.
Friday To (UTC hour) FridayTo	24	Exclusive Friday blocked-window ending hour. 24 means through the end of Friday.
Block Sunday BlockSunday	On	Blocks starting baskets during the Sunday UTC window; does not close existing exposure.
Sunday From (UTC hour) SundayFrom	0	Inclusive Sunday blocked-window starting hour in UTC.
Sunday To (UTC hour) SundayTo	24	Exclusive Sunday blocked-window ending hour. 24 means through the end of Sunday.
Enable Email Alerts EnableEmail	Off	Off in the study. Requires configured platform email and non-empty sender/recipient fields.
Email From EmailFrom	(empty)	Sender address for optional email alerts; empty in the study.
Email To EmailTo	(empty)	Recipient address for optional email alerts; empty in the study.
Email Min Level EmailMinLevel	Warning	Email severity floor; inactive while email alerts are off.
Enable Sound Alerts EnableSound	Off	Off in the study. Sound alerts are live-only.
Sound Min Level SoundMinLevel	Warning	Sound severity floor; inactive while sound alerts are off.
Enable Telegram EnableTelegram	Off	Off in the study. Optional outbound alerts on desktop real-time instances only.
Bot Token TelegramBotToken	(empty)	Private notification token; empty in the study. Native parameter files and reports can retain it.
Chat ID TelegramChatId	(empty)	Notification destination; empty in the study. Redact before sharing files.
Alert Level TelegramMinLevel	Warning	Telegram severity floor; restart after changing notification settings.

> Session windows use UTC, with From inclusive and To exclusive. They prevent starting baskets; they do not close existing exposure. Added-leg guards are off in the study. All notification channels are disabled; native parameter files and reports can retain credentials.

READING THE METRICS

Use the right measure for the question

Terms used throughout the study and the most common interpretation errors.

Term	Meaning in this guide
Net profit / return	Reported net trading profit in USD; return is cumulative over the full test period, not an annual or monthly return.
Balance / equity	Balance reflects realized activity. Equity also includes current open-position P/L; historical equity drawdown can therefore be far larger.
Maximum drawdown	The largest observed peak-to-trough decline for the named series. Equity, balance and resampled closed-trade drawdown describe different series.
Trade / basket	A trade is a closed grid leg. A basket is a managed group of same-direction legs. One profitable basket may contain losing legs.
Profit factor / win rate	Profit factor is gross profit divided by absolute gross loss. Win rate is the share of profitable closed trades. Neither describes the size of all potential future losses.
Completed / end-of-run basket	Completed-basket statistics omit baskets classified EndOfRun. 4 runs have one such basket; its P/L remains in total profit even when the final account is flat.
P5 / median / P95	The fifth, fiftieth and ninety-fifth percentiles of resampled net profit. These describe the finite model sample, not a guaranteed future interval.
VaR95 / CVaR95	In these exports, lower-tail net-profit statistics: P5 profit and average lower-tail profit. A positive figure is not a maximum-loss allowance.
Floor breach (“ruin”)	A resampled path reaching the absolute USD 700 equity threshold, equal to 70% of the USD 1,000 starting equity used in all thirteen simulations.

Do the bucket figures describe one shared account? No. Pages 43–44 combine ending wealth only for separately funded USD 1,000 sleeves. Shared-account results require synchronized exposure, common balance-based risk rules, margin and chronological combined equity.

EVIDENCE AND LIMITATIONS

A documented snapshot, with clear boundaries

A complete thirteen-pair dataset, checked against its preserved native reports and Monte Carlo sources.

Authoritative records

The authoritative source is **Resultados-backtesting-13-pares-actualizado.xlsx**. It is byte-for-byte identical to Resultados-backtesting.xlsx and the preserved completed-study copies. All thirteen rows are actual USD 1,000 backtests; their Monte Carlo sources use the same USD 700 floor.

All thirteen historical report hashes and thirteen Monte Carlo hashes, contexts and headline metrics were checked. The thirteen historical captures match the supplied files. Monte Carlo bands are redrawn from each matched JSON source over normalized trade progress; screenshot calendar axes are not treated as future dates. Numerical claims use the preserved native reports and JSON records.

Evidence boundary	Treatment in this edition
Complete panel	Thirteen historical tests and thirteen matched Monte Carlo sources. No pending result is represented as zero.
Updated records	USDJPY uses the real USD 1,000 rerun. EURCHF uses its completed 267-trade record; NZDUSD and EURCAD now have matching USD 700-floor Monte Carlo sources.
Monte Carlo provenance	USDJPY, EURCHF, EURCAD and CADCHF use local cTrader-engine calculations. The other nine sources are original interface exports. These are distinct source types.
Execution errors	8 logs contain 278 critical EntityNotFound messages: USDCAD 86; USDCHF 39; USDJPY 128; AUDCHF 6; EURAUD 9; EURCHF 4; EURGBP 5; EURCAD 1. Finished status does not resolve the errors.
Costs and spread	Commission is recorded as USD 30 per million USD volume. Native spread is unreported; the 2-pip strategy guard does not establish a fixed backtest spread.

Simulation and product limits

These are historical simulations with hindsight, not live results, an independent audit or evidence from untouched out-of-sample data. They cannot reproduce every liquidity, gap, slippage, outage or broker failure scenario. No expected return, account allocation or guaranteed loss ceiling is implied.

Documentation references: cTrader installation · Saved parameter files · Backtesting. Historical evidence retains the v2.2.2 source context. Current 2.3 source removes ADX and passes 126 protective-execution checks; no new native 2.3 study was run. Platform references accessed 15 October 2026.

Source workbook SHA-256:
7e236a95308b8a3ccb55a215950ae803fbfa22a28a769668cfa85af8399bfb9a

SOURCES / FURTHER READING

Where the operating guidance comes from

Local product evidence and research references serve different purposes. Earlier manuals do not replace the current source or the thirteen-pair study.

Source	Use in this edition
Current robot source	Product-Release / 2.3 / revisions / 20261018-no-adx: complete strategy, protection, recovery, execution, margin-state and alert modules. Current package: SGU Smart Grid Ultimate.algo, assembly 2.3.0.0 / 2.3.0-no-adx.2. The AB A study source remains historical 2.2.2.
Version and operating notes	Retained v2.2.0-2.3 notes plus the 2.3 release notes, source manifest and validation record. The 2.3 release record supersedes earlier package and ADX instructions.
Original SGU manual	SGU-v2.1-Product-Documentation.pdf, PDF pp. 20–21, Calibration Protocol. Source for the limited-search principle. Older result panels and categorical claims about sample sufficiency are not reused.
López de Prado, 2018	Advances in Financial Machine Learning, local AFML - Lopez de Prado.pdf. Ch. 7, printed pp. 104–109 (PDF 131–136): overlap; Ch. 11, pp. 151–157 (PDF 178–184): testing and selection; Ch. 12, pp. 161–162 (PDF 188–189): validation.
Bailey et al., 2015	The Probability of Backtest Overfitting, revised February 2015. Explains selection across repeated backtests and proposes an assessment framework; it does not evaluate SGU.
Companion material	GIL2476_AdvancesFinancial_BonusPDF.pdf, PDF pp. 65, 94–97: supporting figures and code on overlap and backtest selection. Companion to the book, not independent evidence of SGU performance.
Local optimization study	Estudio-Optimizacion / Presets-H1 and archived EURUSD manifest 20261014T173114Z-ConfirmTF-K21. Useful procedures and limitations; its USD 500 settings, dates and assumptions differ from the published panel.

Current release files were reviewed under **Backtesting-SGU / Product-Release / 2.3**; historical product files under **cAlgo / Sources / Robots**. The book is in **Fast Tree ML.NET** (a duplicate exists in ML.NET Test); the companion is in **Meta-Labeling-Robot / Meta-Labeling-Robot**. PDF page numbers count from the first file page; printed numbers refer to the book's folios.

Platform procedures: Install · Saved parameters · Backtest · Optimize. Official cTrader documentation checked 15 October 2026.

> The optimization chapter adapts research principles to this robot. It does not report a completed walk-forward, purged cross-validation, probability-of-backtest-overfitting or deflated-Sharpe analysis. No research citation certifies a configuration as safe or profitable.

CRAFIUM.

SMART GRID ULTIMATE

Understand the system.
Verify your setup.
Stay in control.

PRODUCT SUPPORT

cg.walter.ma@gmail.com

+56 9 3366 9343

TO HELP US RESOLVE YOUR REQUEST

Include the build, broker, symbol, timeframe, run ID
and a redacted log or native report.

Use the exact package, symbol, account and settings you have validated.
Confirm exposure and protection directly in cTrader.

Trading foreign exchange on margin involves substantial risk. Historical and resampled results do not guarantee future performance. Protective controls depend on execution and cannot ensure a maximum loss. This guide is technical product information and does not recommend an instrument or allocation.